

Introduction of background knowledge and concepts in biology and medicine

A: Background concepts

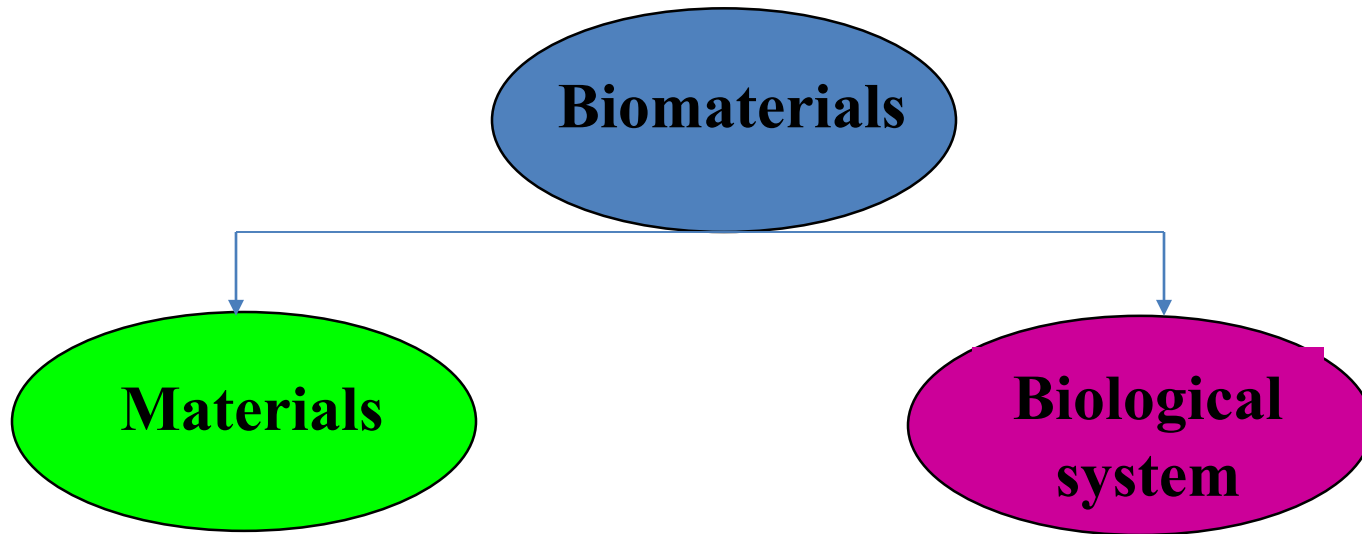
B: Proteins: Structure, Properties, and
Adsorption to Surfaces

D: Cells

E: Extracellular matrix, and
cell-biomaterial interactions

Biomaterial Science

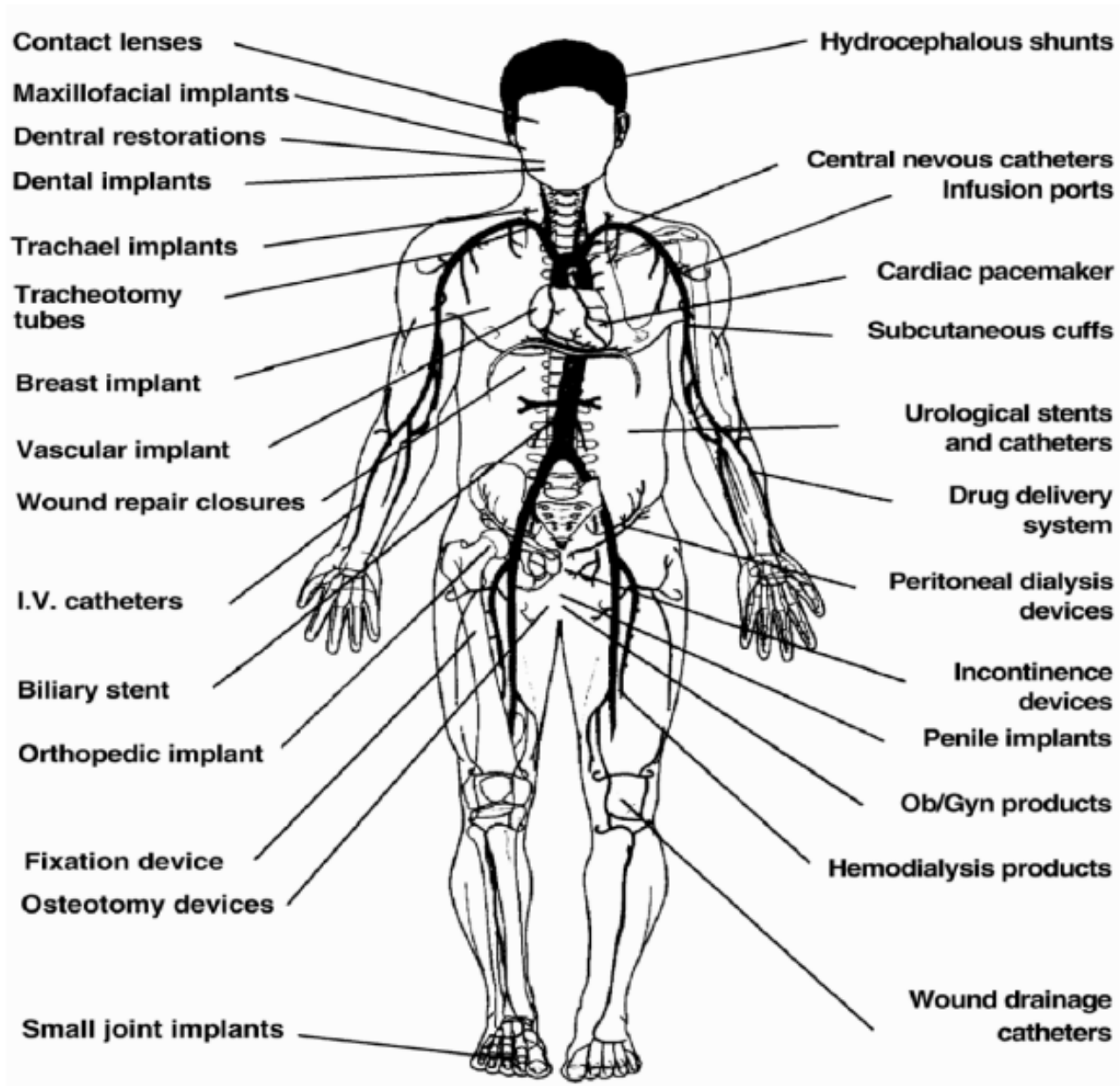
Biomaterial — A biomaterial is a nonviable material used in a (medical) device intended to interact with biological systems (Williams 1987).



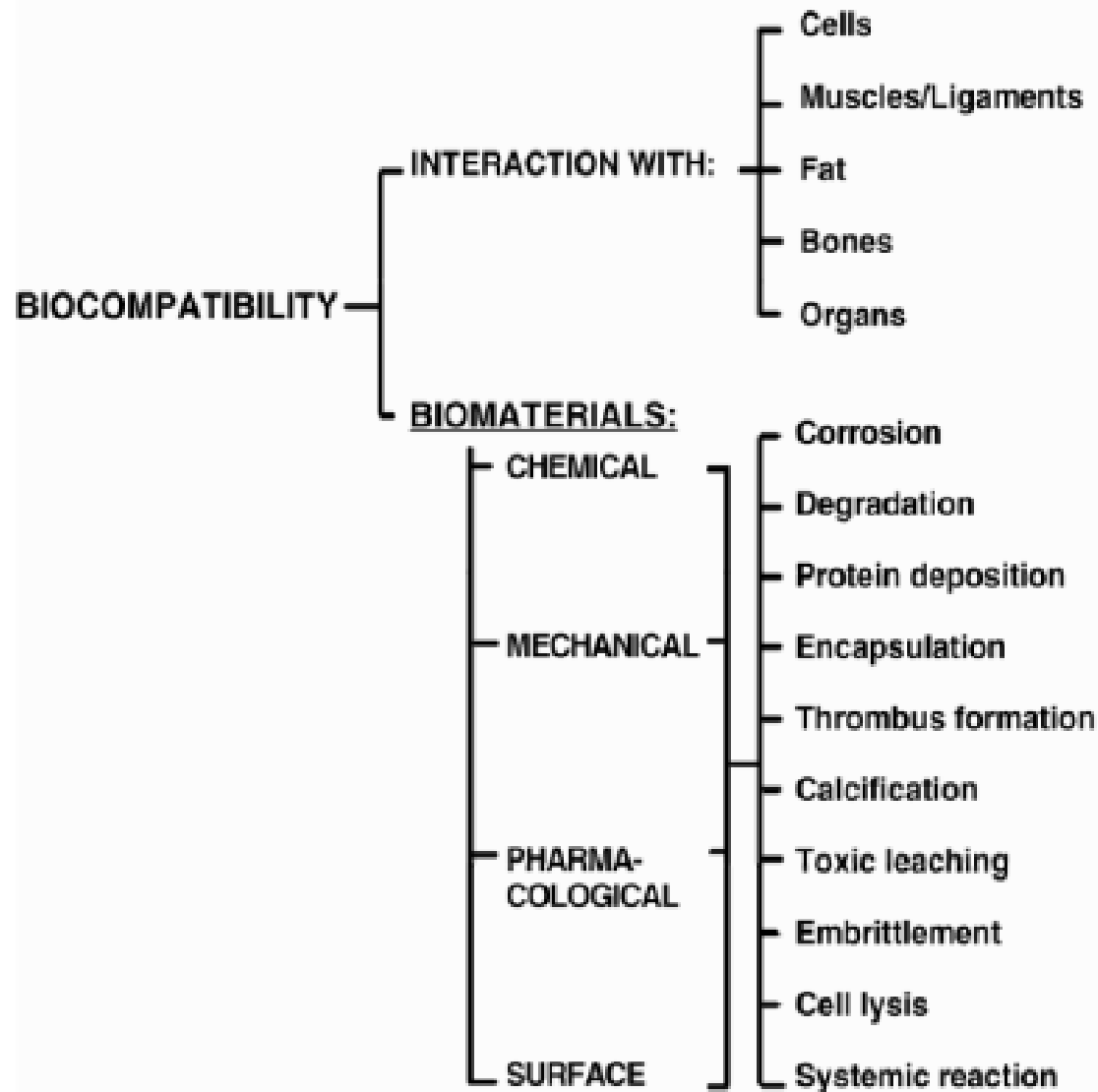
Biocompatibility — The ability of a material to perform with an appropriate host response in a specific application (Williams 1987).

Host Response — The response of the host organism (local and systemic) to the implanted material or device.

Various implants and devices for replacement or enhancement



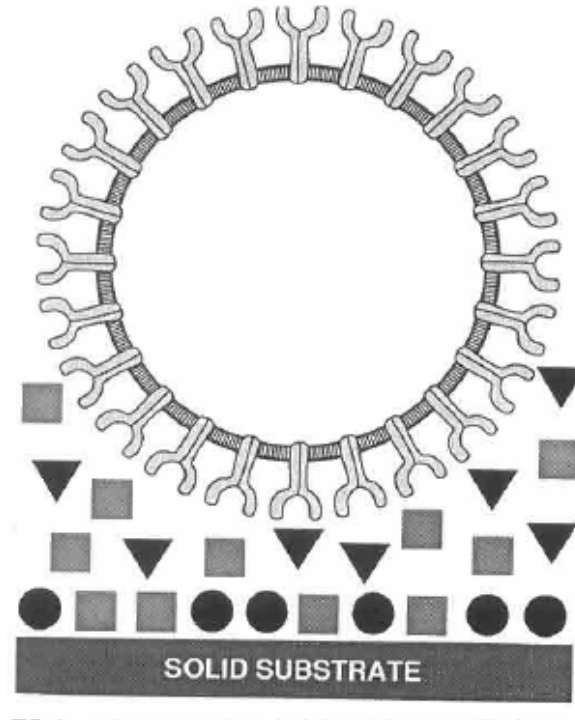
Biocompatibility



A successful biomaterial implant must support all the required functions of the attached (or neighboring) cells, including

- | | |
|------------------------------|-----------------|
| • Viability | All cell types |
| • Communication | All cell types |
| • Protein synthesis | All cell types |
| • Proliferation | Some cell types |
| • Migration | Some cell types |
| • Activation/differentiation | Some cell types |
| • Programmed cell death | Some cell types |

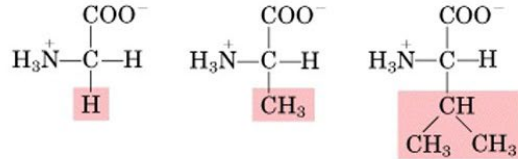
Proteins: Structures, Properties and Adsorption to Surfaces



Cell interaction with an adsorbed protein layer on a solid substrate. The cell is shown as circular space with a bilayer membrane in which the adhesion receptor protein molecules (the slingshot-shaped objects) are partly embedded. The proteins in the extracellular fluid are represented by squares and triangles.

Structure and Properties of Amino Acid Side Chains

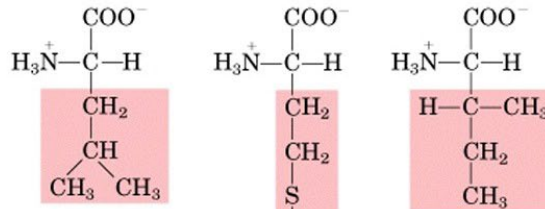
Nonpolar, aliphatic R groups



Glycine

Alanine

Valine

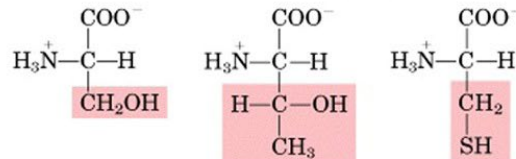


Leucine

Methionine

Isoleucine

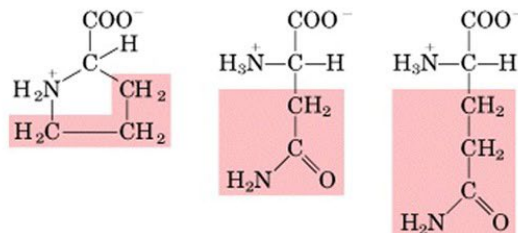
Polar, uncharged R groups



Serine

Threonine

Cysteine

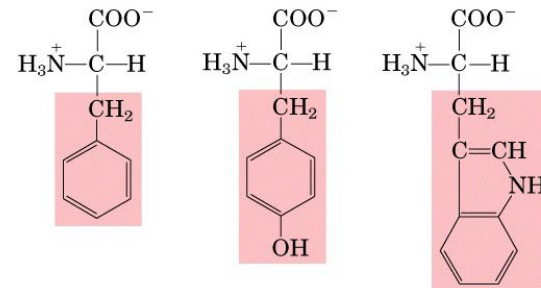


Proline

Asparagine

Glutamine

Aromatic R groups

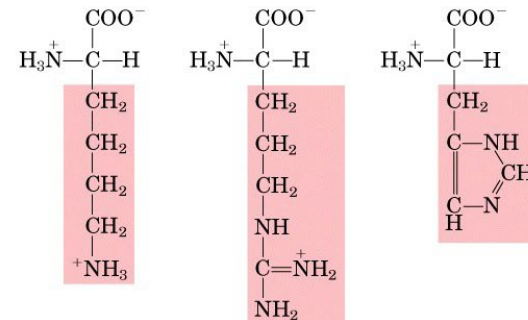


Phenylalanine

Tyrosine

Tryptophan

Positively charged R groups

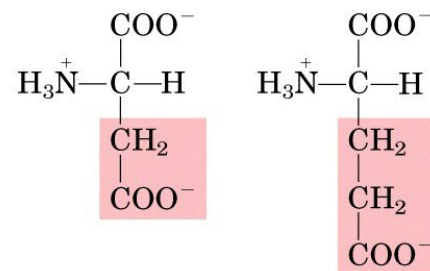


Lysine

Arginine

Histidine

Negatively charged R groups



Aspartate

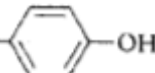
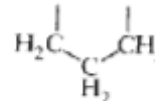
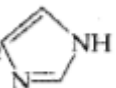
Glutamate

Collagen

key
amino
acids?

Structure and Properties of Amino Acid Side Chains

TABLE 1 Structure and Properties of Amino Acid Side Chains

Amino acid	Structure	Group and pK	Charge	Hydrophobicity ^a	Surface tension ^b
Methionine	$-\text{CH}_2-\text{CH}_2-\text{S}-\text{CH}_3$		Neutral	0.26	-3.01
Alanine	$-\text{CH}_3$		Neutral	0.25	0.96
Glycine	$-\text{H}$		Neutral	0.16	1.12
Cysteine	$-\text{CH}_2\text{SH}$	$-\text{SH}$: 8.3	0 to -1	0.04	0.69
Tyrosine	$-\text{CH}_2-$ 	$-\text{OH}$: 10.9	0 to -1	0.02	-15.1
Proline			Neutral	-0.07	-0.49
Threonine	$-\text{CH}-\text{OH}$ $ $ CH_3		Neutral	-0.18	0.59
Serine	$-\text{CH}_2-\text{OH}$		Neutral	-0.26	0.76
Histidine	$-\text{CH}_2-$ 	$-\text{NH}-$: 6.0	0 to +1	-0.40	1.03
Glutamic acid	$-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$	$-\text{CO}_2\text{H}$: 4.3	0 to -1	-0.62	0.86
Asparagine	$-\text{CH}_2-\text{C}(=\text{O})-\text{NH}_2$		Neutral	-0.64	1.17
Glutamine	$-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})-\text{NH}_2$		Neutral	-0.69	1.21
Aspartic acid	$-\text{CH}_2-\text{CO}_2\text{H}$	$-\text{CO}_2\text{H}$: 3.9	0 to -1	-0.72	0.96
Lysine	$-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{NH}_3^+$	$-\text{NH}_2$: 10.8	0 to +1	-1.1	0.92
Arginine	$-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{NH}-\text{C}(=\text{NH}_2)-\text{NH}_2$	$-\text{NH}_2$: 12.5	0 to +1	-1.8	1.03

Structure and Properties of Amino Acid Side Chains

TABLE 1 Structure and Properties of Amino Acid Side Chains

Amino acid	Structure	Group and pK	Charge	Hydrophobicity ^a	Surface tension ^b
Isoleucine	$\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}-\text{CH}_2-\text{CH}_3 \end{array}$		Neutral	0.73	-15.2
Phenylalanine	$-\text{CH}_2-\text{C}_6\text{H}_5$		Neutral	0.61	-17.3
Valine	$\begin{array}{c} -\text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		Neutral	0.54	-3.74
Leucine	$\begin{array}{c} -\text{CH}_2\text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		Neutral	0.53	-21.9
Tryptophane	$-\text{CH}_2-\text{C} \begin{array}{l} \text{---} \text{Indole ring} \\ \parallel \\ \text{HC} \\ \\ \text{N} \\ \\ \text{H} \end{array}$		Neutral	0.37	-9.6
Methionine	$-\text{CH}_2-\text{CH}_2-\text{S}-\text{CH}_3$		Neutral	0.26	-3.01
Alanine	$-\text{CH}_3$		Neutral	0.25	0.96
Glycine	$-\text{H}$		Neutral	0.16	1.12
Cysteine	$-\text{CH}_2\text{SH}$	$-\text{SH}: 8.3$	0 to -1	0.04	0.69
Tyrosine	$-\text{CH}_2-\text{C}_6\text{H}_4-\text{OH}$	$-\text{OH}: 10.9$	0 to -1	0.02	-15.1

Properties of Selected Proteins

TABLE 2 Properties of Selected Proteins

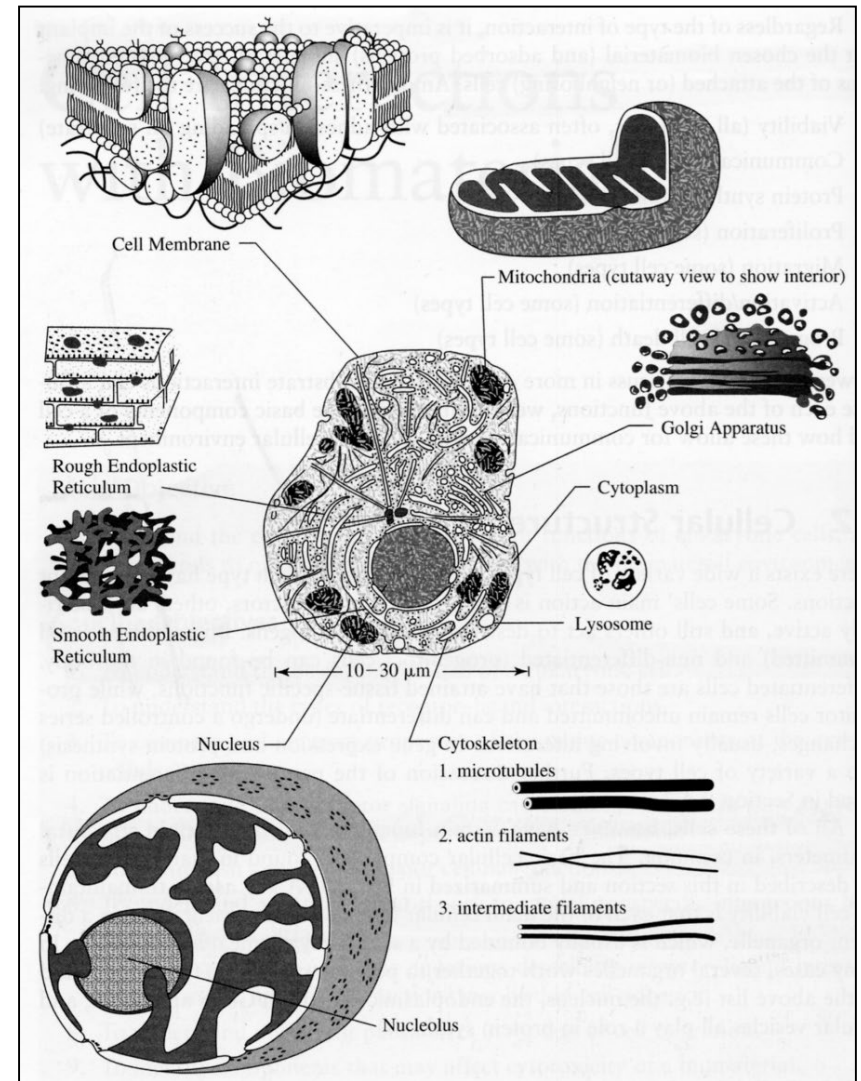
Protein	Function	Location	Size	Shape ^a	pI	Stability ^a	Surface activity
Albumin	Carrier	Blood	65 kDa	42 × 141 Å (1)	4.8	Denatures at 60°C (8)	Low on PE
Fibrinogen	Clotting	Blood	340 kDa	460 × 60 Å (2) trinodular string	5.8	Denatures at 56°C (9)	High on PE
IgG	Antibody	Blood	165 kDa	T-Shaped	6.5		Low on PE
Lysozyme	Bacterial lysis	Tear; hen egg	14.6 kDa	45 × 30 × Å (3) Globular	11	$\Delta G_n = -14$ kcal/mol (10)	High on negatively charged surfaces
Hemoglobin	Oxygen carrier	Red cells	65 kDa	55 Å (4) Spherical A2B2 tetramer	6.87	Normal form	Very high on PE
Hemoglobin S	Oxygen carrier	Sickle red cells	65 kDa	55 Å (4) Spherical A2B2 tetramer	7.09	Lower than A form	Oxy form of HbS has much higher air-water activity than normal Hb
Myoglobin	Oxygen carrier	Muscle	16.7 kDa	45 × 35 × 25 Å (5) Spherical monomer	7.0	$\Delta G_n = -12$ kcal/mol (10)	Unknown
Collagen	Matrix factor	Tissue	285 kDa	3000 × 15 Å (6) Triple helical rod		Melts at 39°C (11)	
Bacterio-rhodopsin	Membrane protein		26 kDa	30–40 Å long (7) Seven-rod structure that self-localizes in membranes			High at cell membrane
Tryptophan synthase alpha subunit ("wild type")	Enzyme		27 kDa		5.3	$\Delta G_n = -8.8$ kcal/mole; denatures at 55°C (12)	High air-water activity compared with ovalbumin
Tryptophan synthase (glu → ileu) variant alpha subunit	Enzyme		27 kDa			$\Delta G_n = -16.8$ kcal/mole (12)	Much less active at air-water interface than wild type

Cell Structure

2 types of cells:

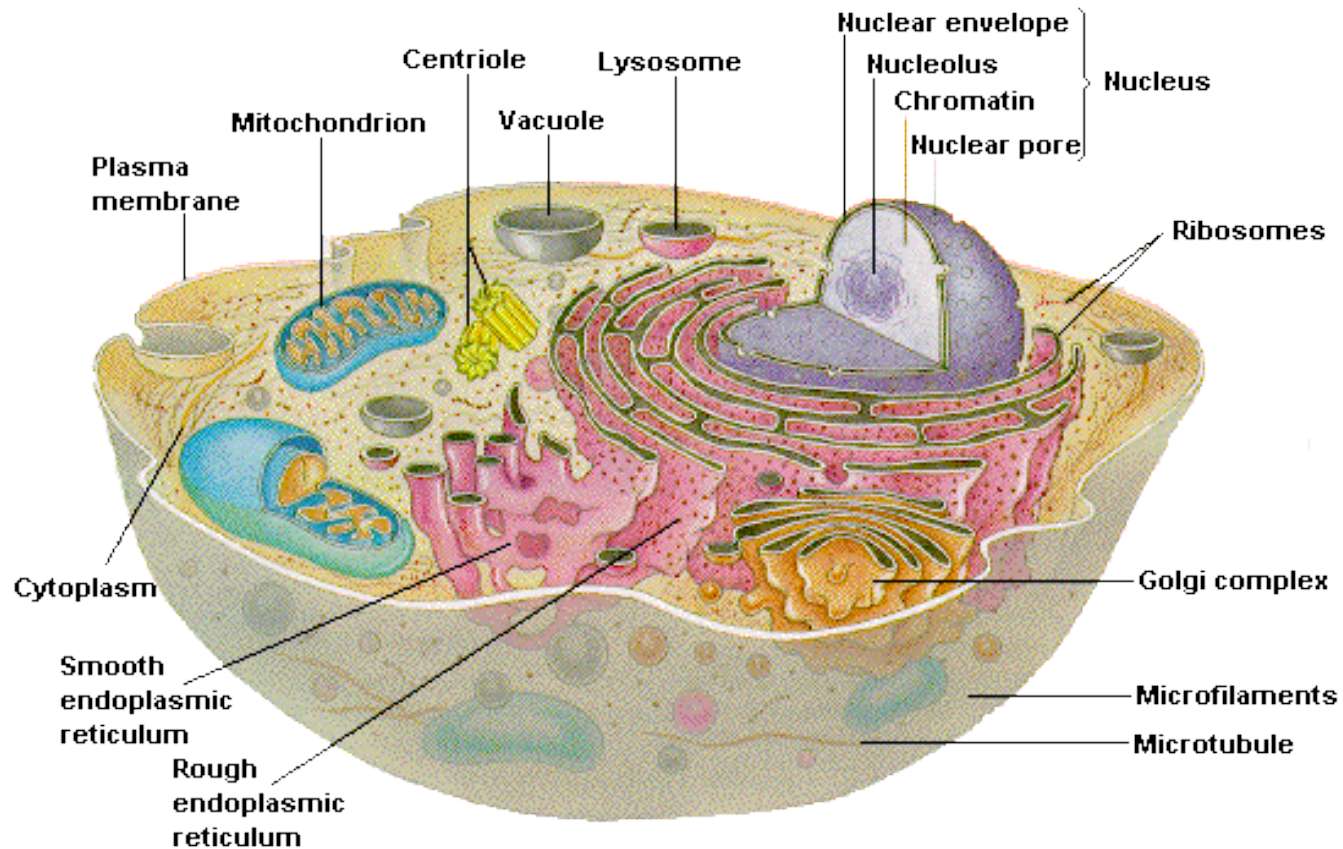
- Differentiated: perform specific tissue functions
- Undifferentiated: progenitors for many different cell types

Cell tasks are compartmentalized in various **organelles**. Organelles in all mammal cells include the plasma cell membrane, mitochondria, Golgi apparatus, cytoplasm, lysosome, cytoskeleton, nucleus, and smooth and rough endoplasmic reticulum



细胞在彻底分化前，能转化成某种中间细胞（intermediate cell），这种中间细胞被称作祖细胞（progenitor cell）或前体细胞（precursor cell）。祖细胞属于成体干细胞，是未分化的多能或专能干细胞。

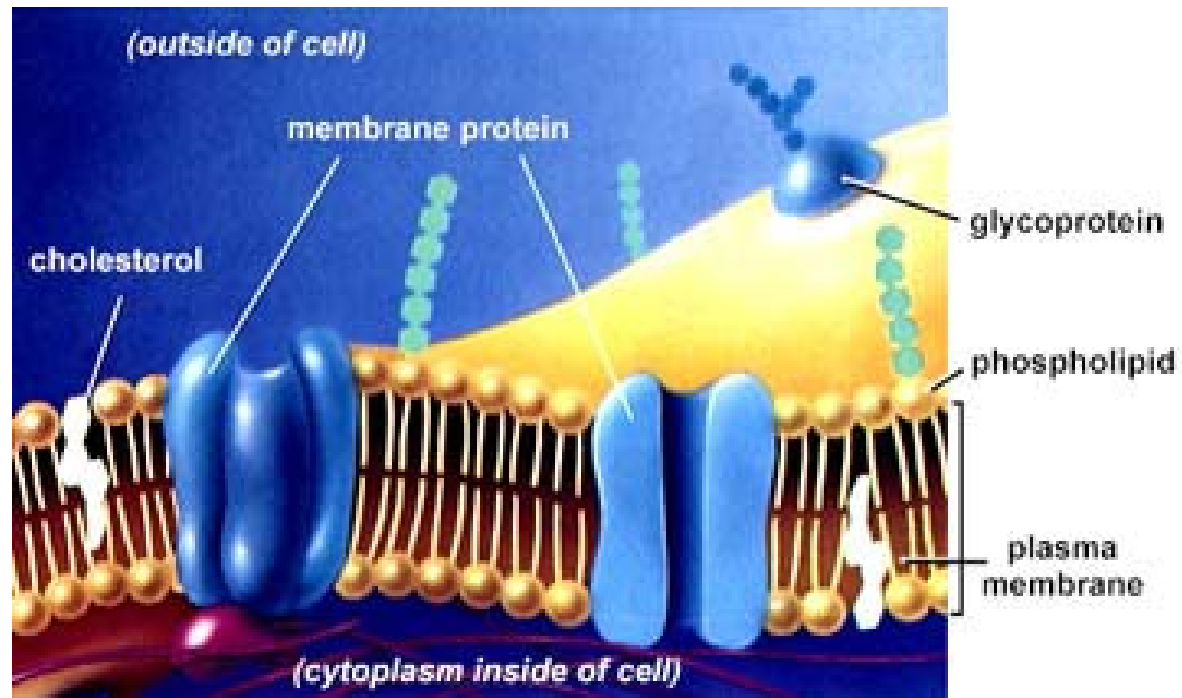
Cell Structure



Cell tasks are compartmentalized in various ***organelles***.

Organelles in all mammal cells include the plasma cell membrane, mitochondria, Golgi apparatus, cytoplasm, lysosome, cytoskeleton, nucleus, and smooth and rough endoplasmic reticulum.

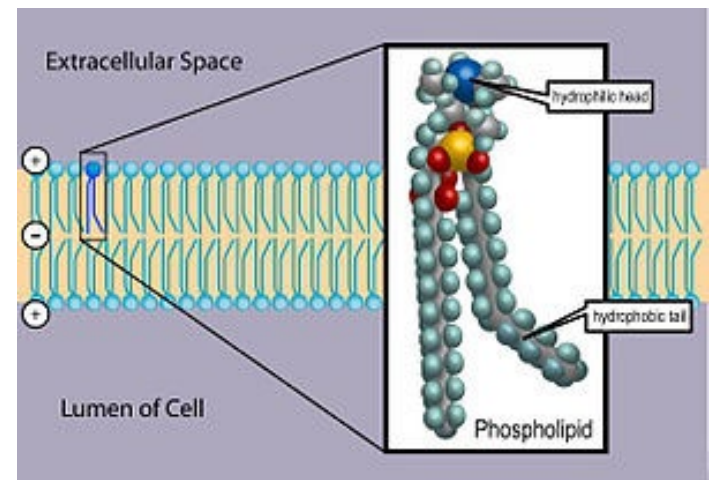
Cell membrane



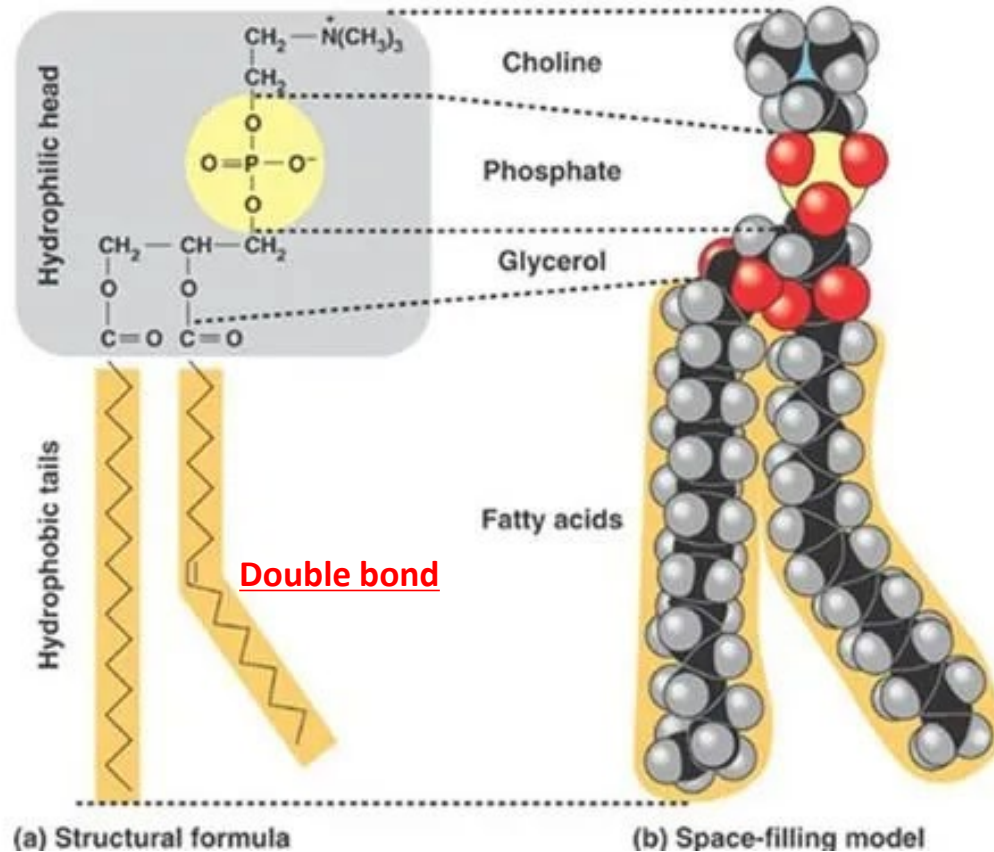
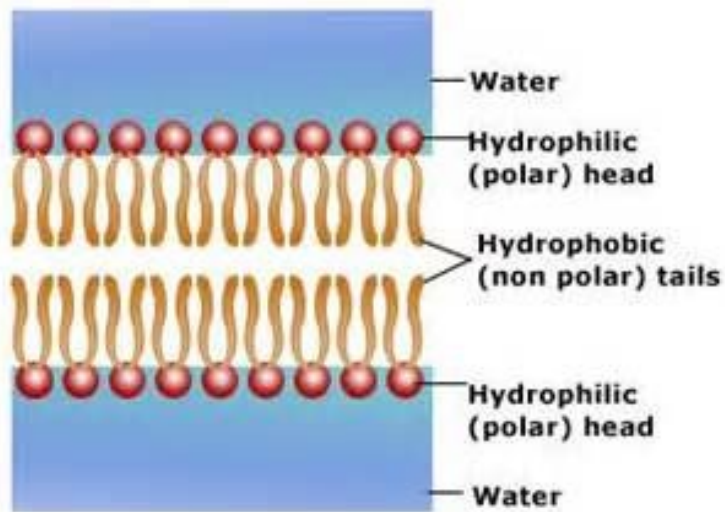
The **cell membrane** separates the **cytoplasm**, or cell interior, from the aqueous external environment. It is a bilayered structure made up of **phospholipids**, or fatty acids with a polar (hydrophilic) head and nonpolar tail.

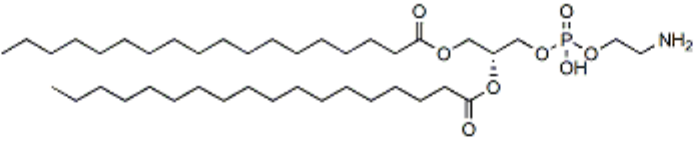
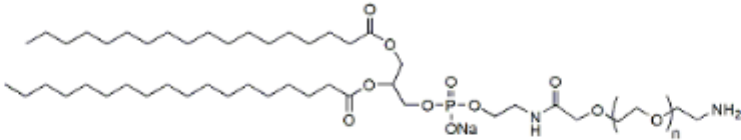
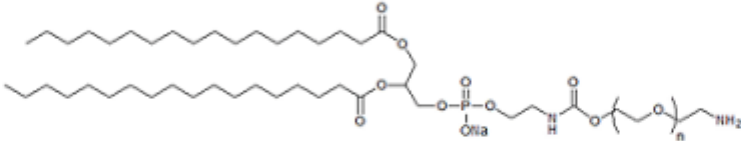
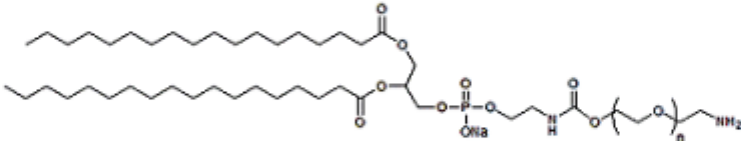
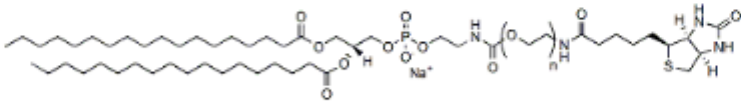
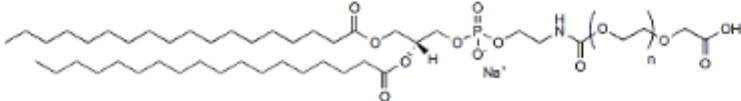
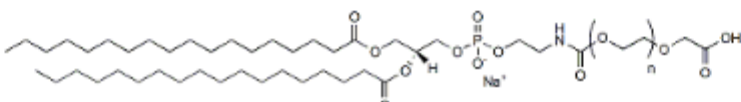
Transmembrane proteins span the cell membrane to channel ions into and out of cell and maintain proper cell chemistry

Other proteins target specific extracellular molecules

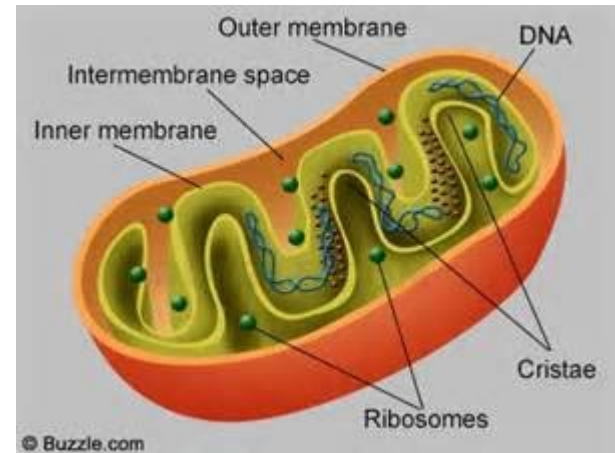
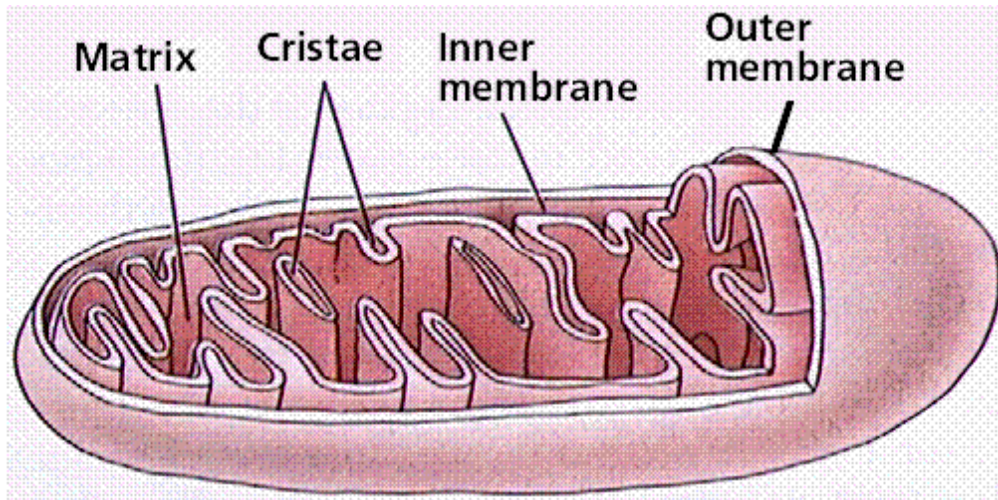


Phospholipid

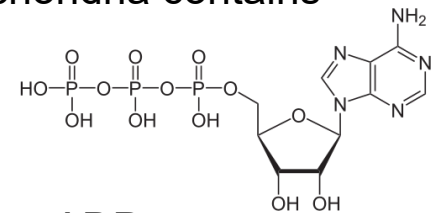


BP-23601	1,2-DISTEAROYL-SN-GLYCERO-3-PHOSPHOETHANOLAMINE		748.1	95%	Pricing
BP-22722	DSPE-PEG-2K-Amine				Pricing
BP-23200	DSPE-PEG-3.4K-Amine				Pricing
BP-23694	DSPE-PEG-5K-Amine				Pricing
BP-22723	DSPE-PEG-2K-Biotin				Pricing
BP-22719	DSPE-PEG-2K-CH2COOH				Pricing
BP-23703	DSPE-PEG-5K-CH2COOH				Pricing

Mitochondria

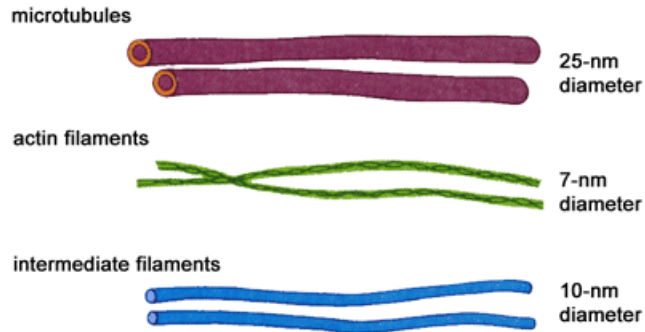


The **mitochondria** produce the energy for cell functions via a process called **oxidative phosphorylation**. Surrounded by a phospholipid membrane, the mitochondria contains **enzymes** that help break down molecules



In oxidative phosphorylation ATP (adenosine triphosphate) is converted to ADP (adenosine diphosphate, an exothermic reaction that releases energy to drive the cellular process.

Cytoskeleton and Golgi

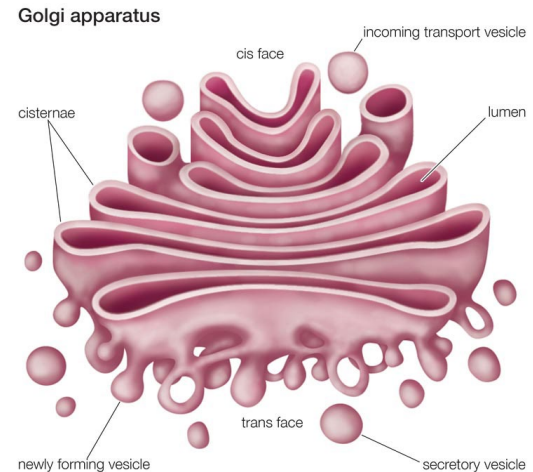


The **cytoskeleton** is made up of three filaments:

- Actin microfibrils, ~6-8 nm diameter
- Intermediate filaments ~10 nm diameter
- Microtubules ~25 nm diameter

Made up of proteins, the **cytoskeleton**

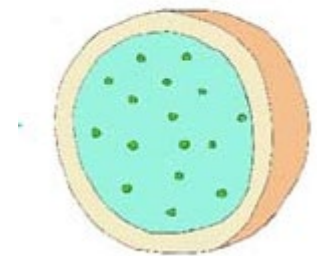
- Gives the cell shape
- Can provide cell locomotion
- Aids in separation/duplication of DNA



The **Golgi apparatus** modifies, sorts and packages proteins for their final destination

Lysosome

Lysosomes are specialized vesicles with digestive enzymes



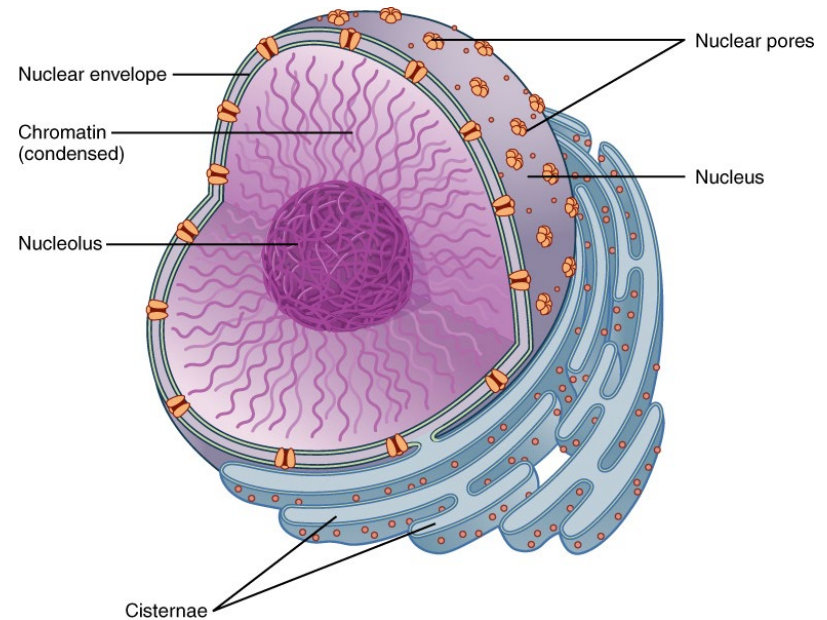
Nucleus

The **nucleus** is the control center for the cell. It contains:

DNA – (deoxyribonucleic acid) that is condensed into **chromatin**, a complex combination of DNA and protein that makes up chromosomes

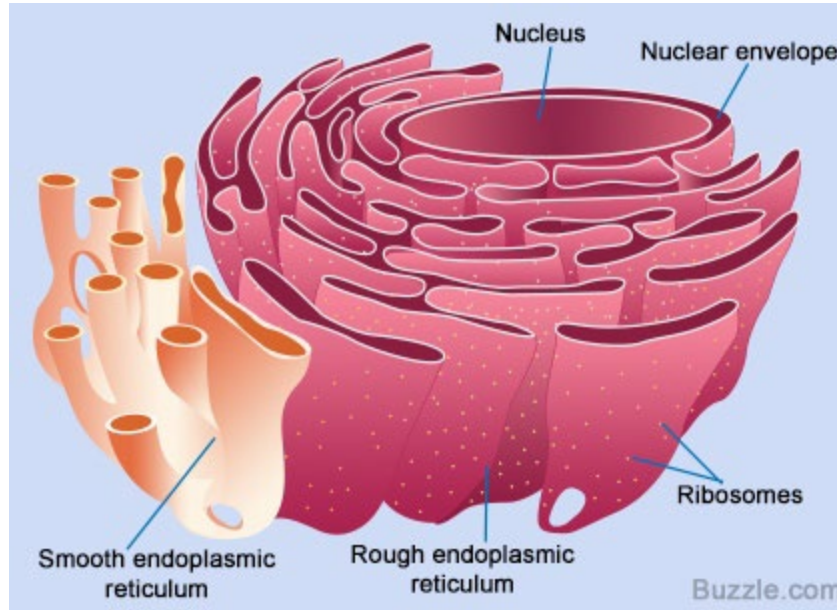
The **nuclear envelope**, a bilayer of phospholipid membranes that surrounds the nucleus

The outer membrane of the nuclear envelope is contiguous with the endoplasmic reticulum and is connected with the inner membrane at specific locations called **nuclear pores**. The pores are composed of proteins that form gates to allow only specific molecules in and out of the nucleus



Ribosomes, minute round particles composed of RNA and protein that are found in the cytoplasm of living cells and catalyze reactions in which mRNA is used to synthesize proteins. *Ribosomes* are located in the **nucleolus**

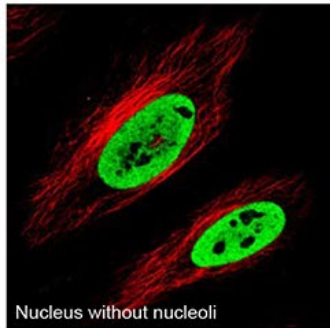
Endoplasmic Reticulum (内质网)



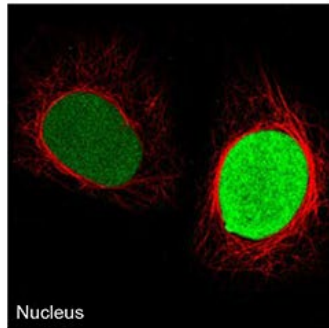
The outer membrane of the nucleus is connected to the ***endoplasmic reticulum (ER)***. The *ER* is the site of protein synthesis. The ER is made of long, **flattened sheets of phospholipids**, and may be either *rough* or *smooth*.

The *rough ER* has ribosomes attached to the surface that act as catalysts for protein synthesis. The *smooth ER* is more tubular and does not contain ribosomes. It packages the proteins produced in the nucleolus and rough ER in phospholipids for delivery to the *Golgi apparatus*.

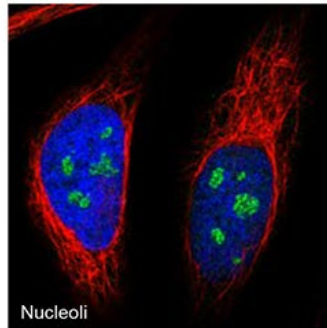
Organelles



Nucleus without nucleoli
IF staining with Anti-FUS (AMAb90549) shows nuclear (without nucleoli) positivity.



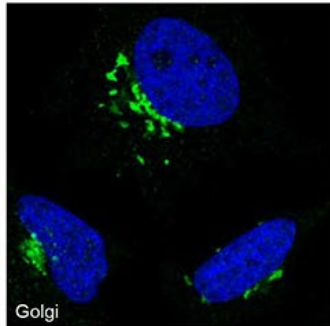
Nucleus
IF staining with Anti-PARP1 (AMAb90959) shows a cell cycle dependent nuclear (without nucleoli) pattern.



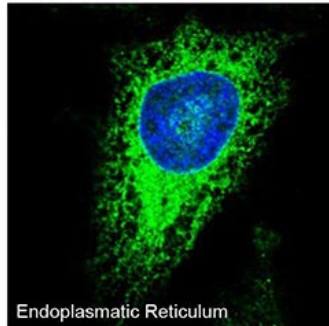
Nucleoli
IF staining with Anti-NOP56 (AMAb91013) shows nucleoli-specific positivity in green.



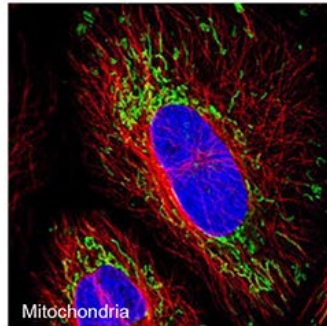
Nuclear membrane
IF staining with Anti-EMD (AMAb90562) shows strong positivity in nuclear membrane, as well as weaker positivity in ER.



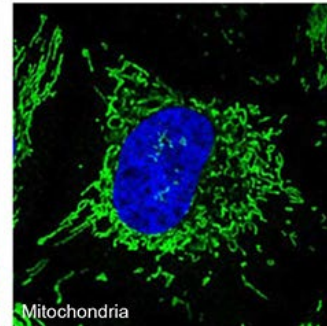
Golgi
IF staining using Anti-GORASP2 (AMAb91016) shows positivity in golgi apparatus.



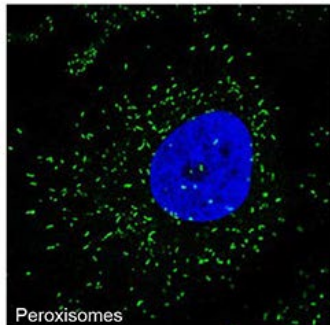
Endoplasmic Reticulum
Anti-PDIA3 (AMAb90988) shows distinct positivity of Endoplasmic Reticulum (ER) using IF.



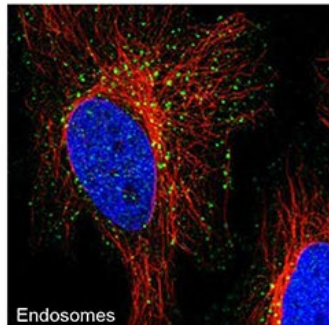
Mitochondria
IF staining using Anti-TUFM (AMAb90966) shows specific mitochondrial staining.



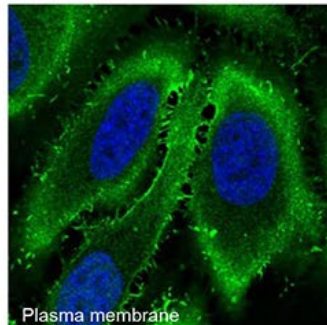
Mitochondria
Anti-CS (AMAb91005) shows distinct positivity in mitochondria.



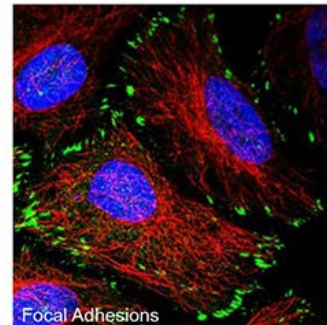
Peroxisomes
The image shows specific staining of peroxisomes using Anti-ABCD3 (AMAb90995).



Endosomes
IF staining with Anti-VPS26A (AMAb90967) shows distinct staining of endosomes.



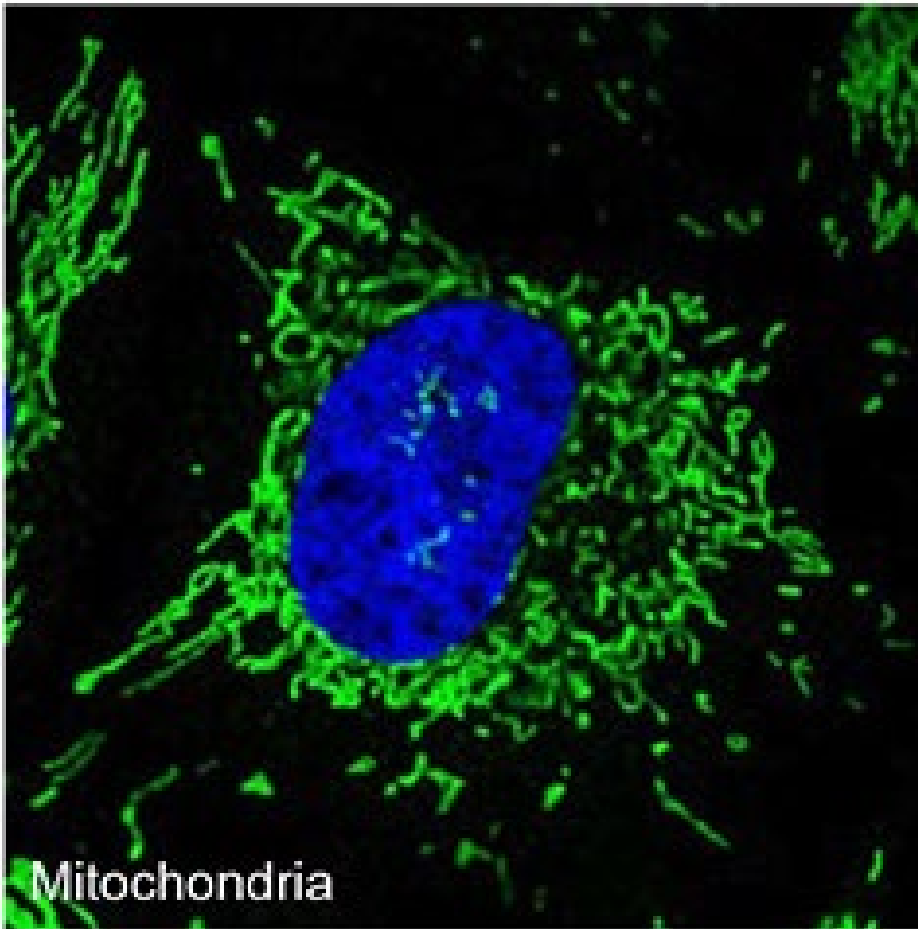
Plasma membrane
The image shows specific plasma membrane staining using Anti-EZR (AMAb90976).



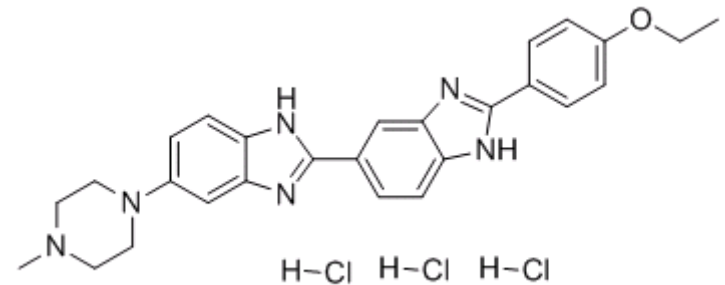
Focal Adhesions
IF staining with Anti-ZYX (AMAb90992) shows distinct staining of focal adhesions.

Immunofluorescence images showing 11 different subcellular organelles in the human HeLa cell line using monoclonal antibodies as markers with antibody staining shown in green. Microtubule- and nuclear probes are visualized in red and blue respectively (where available).

Organelles - Nucleus

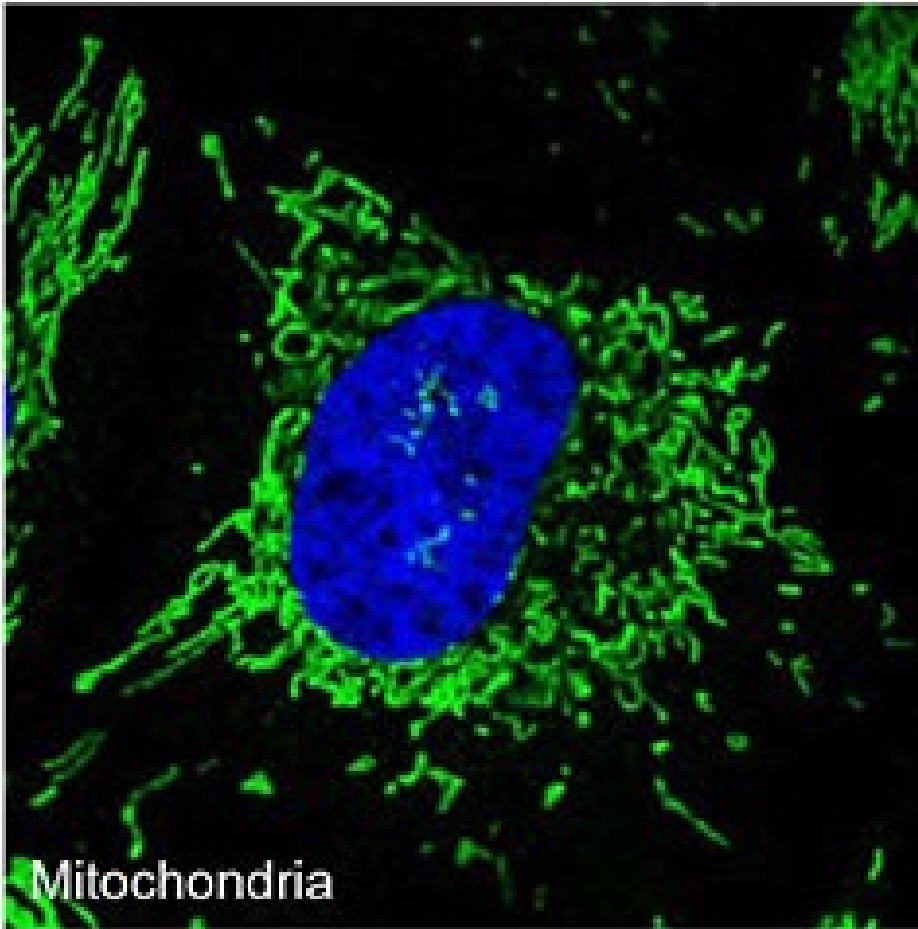


Anti-CS (AMAb91005) shows distinct positivity in mitochondria.



Hoechst 33342 binds to adenine-thymine-rich regions of DNA in the minor groove. On binding to DNA, the fluorescence greatly increases.

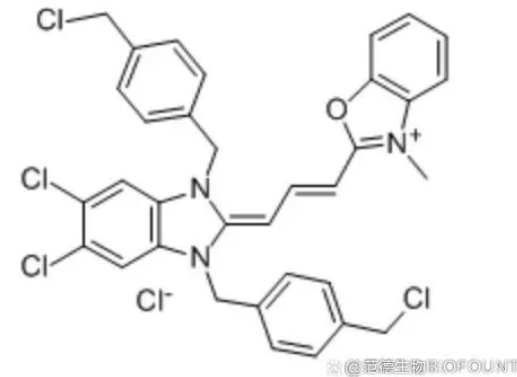
Organelles - Mitochondria



Anti-CS (AMAb91005) shows distinct positivity in mitochondria.

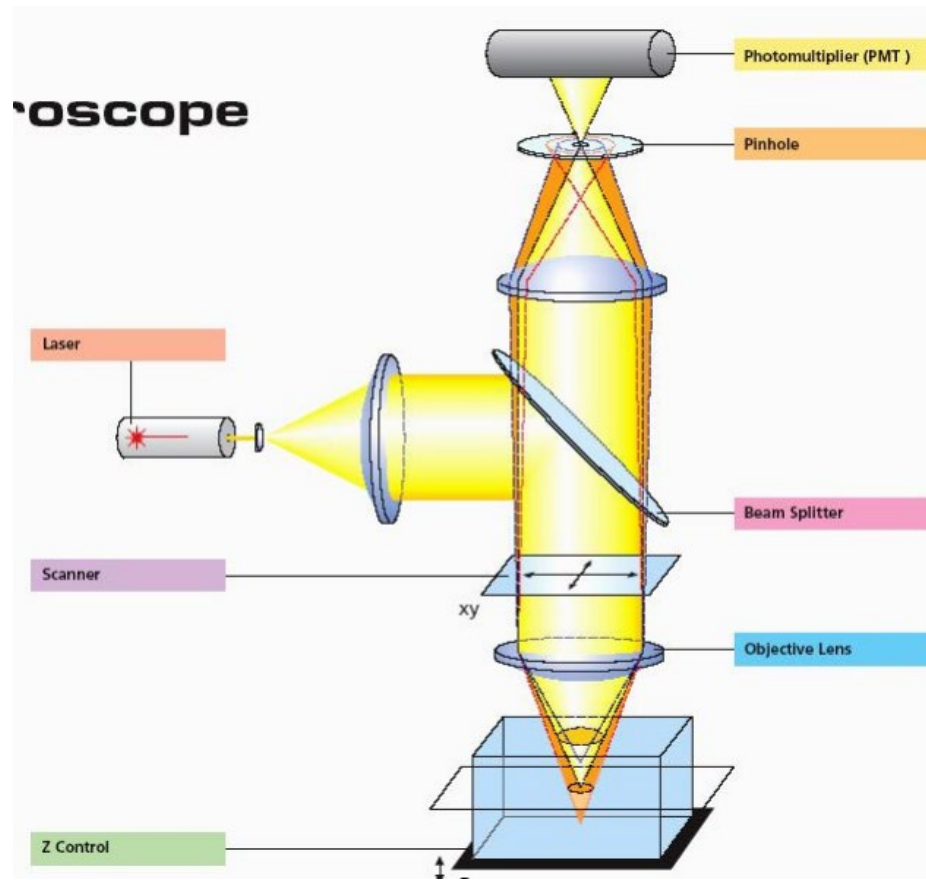
Anti-CS **antibody** AMAb91005

Mitotracker Green FM



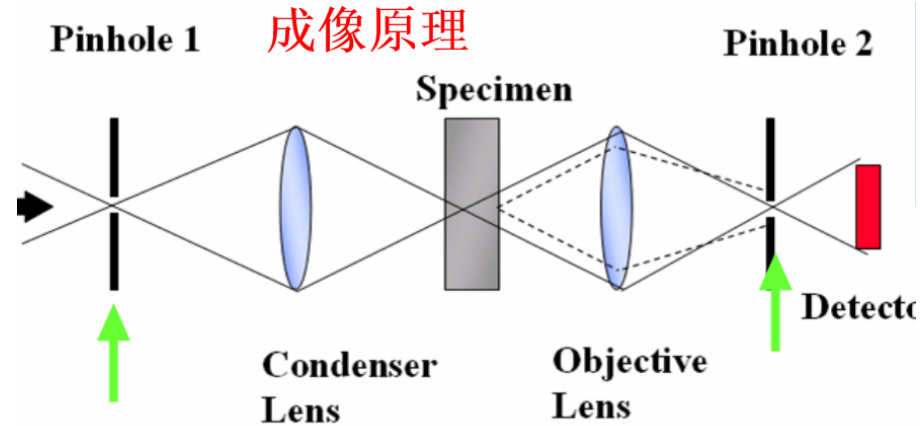
Mitotracker Red???

Brief introduction of confocal microscope



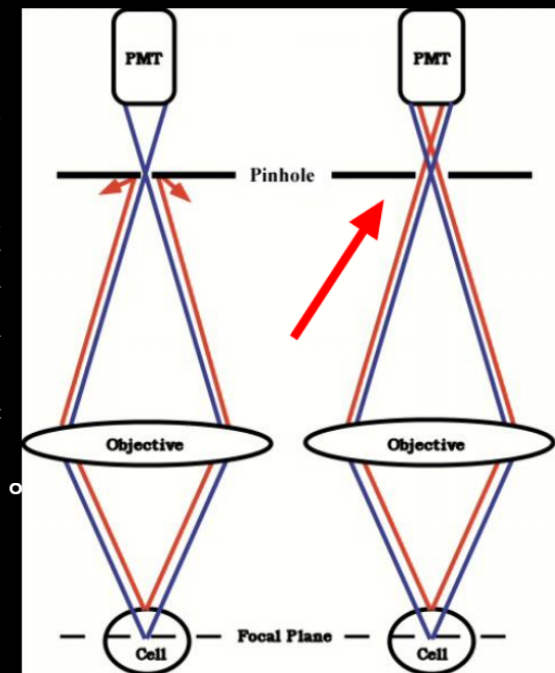
Conditions:
Excitation
Emission
Dyes (fluorophores)
cells

Conditions:
Pinholes
Sample concentrations
Focus
Photobleach



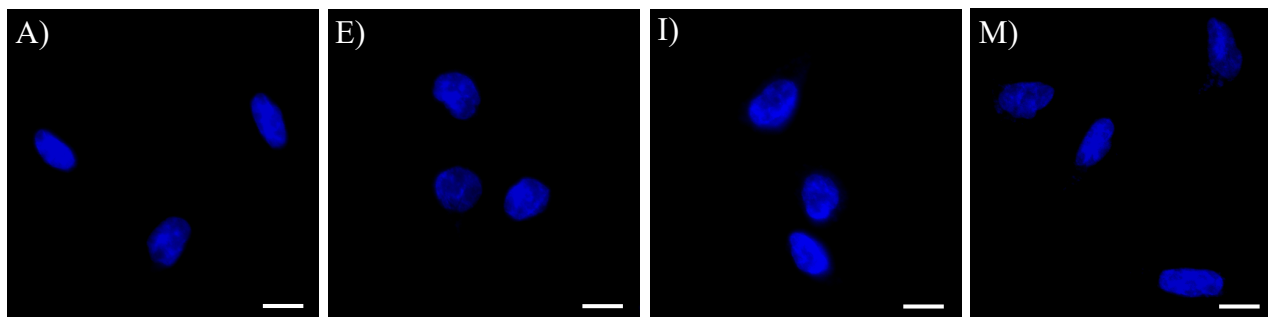
“共聚焦”概念是指光路(激发和发射)在两个位置上聚焦()。在共聚焦扫描仪中,激发光聚焦在样本上而发射光聚焦在针孔上。

只有精确聚焦在样本表面的荧光信号才能通过共聚焦针孔,这样就可以过滤掉非特异的荧光信号。

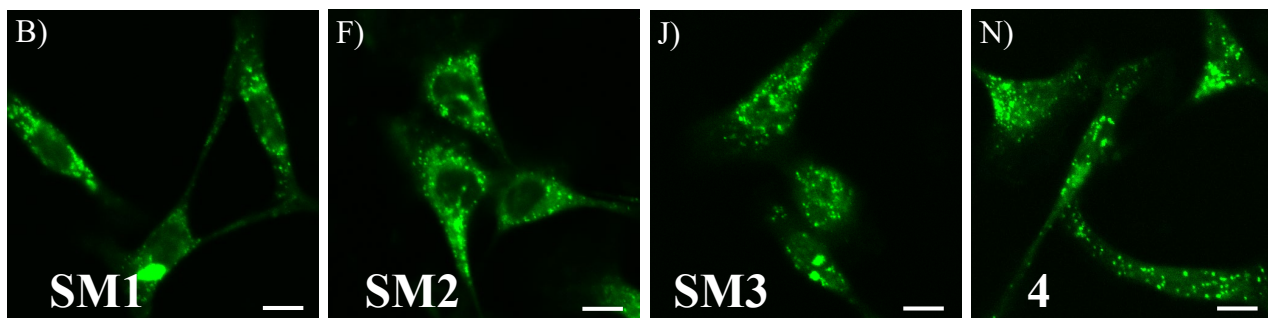


Intracellular pH sensing (CP-A cell line for different materials)

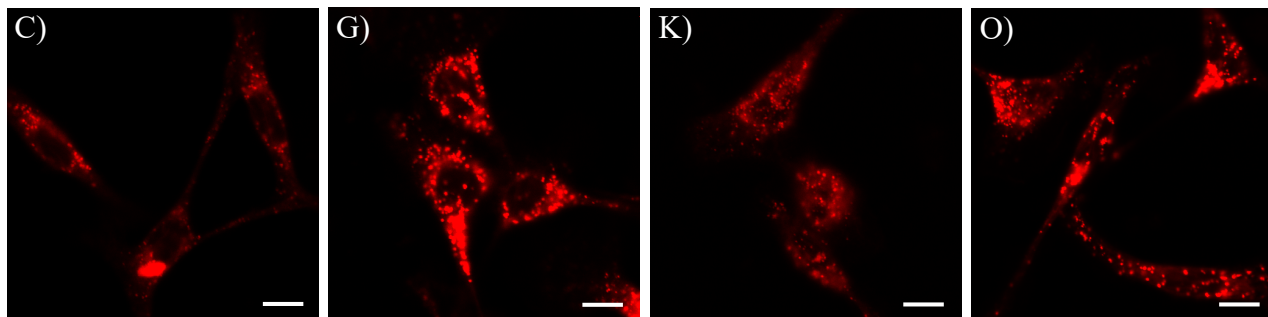
Hoechst 33342



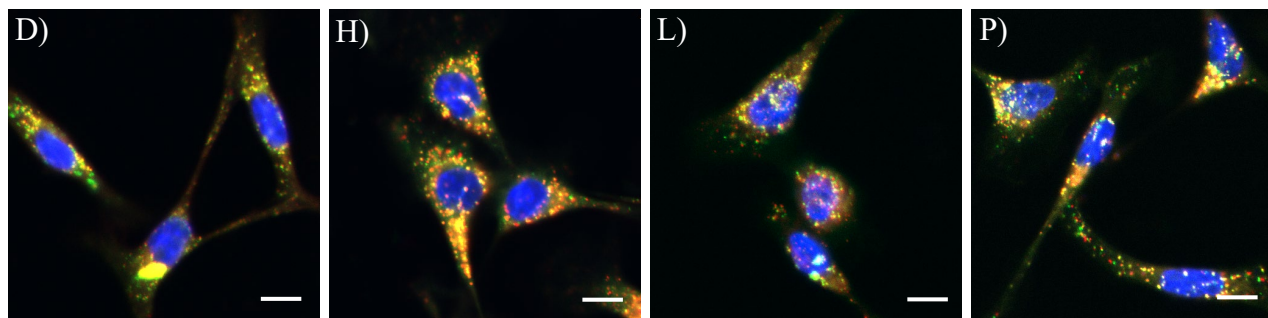
Sensors



LysoTracker Red



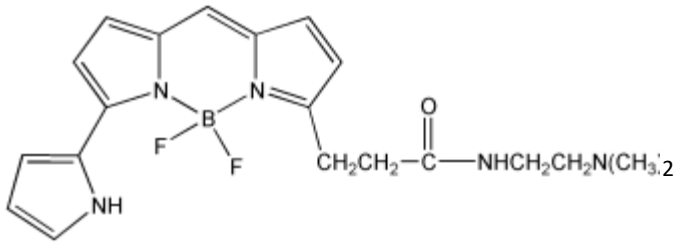
Overlay



One of the best biomaterials related images for year 2010

Biomaterials, 2010

LysoTracker® Red DND-99



溶酶体（lysosomes）一般为真核细胞中的一种细胞器；为单层膜包被的囊状结构，大小（在电镜下显示多为球形，但存在橄榄球形）**直径约0.025~0.8微米**；内含多种水解酶，专为分解各种外源和内源的大分子物质。1955年由比利时学者Cristian de Duve(1917-2013)等人在鼠肝细胞中发现。

In mammalian cells, **organelles work in different pH environments ranging from 4.5 to 8.0.**

Lysosomes, filled with numerous hydrolytic enzymes for cellular hydrolysis exhibit an acidic environment (**pH = 4.5–5.5**)

LysoTracker®系列探针是对活细胞中的酸性区室进行选择染色的一类荧光染料，该类探针具有几大重要的特点，**1）**选择性标记酸性细胞器；**2）**纳摩尔级（nM）浓度即可有效标记活细胞；**3）**具有多色探针提供，可根据情况对样品进行多标实验。

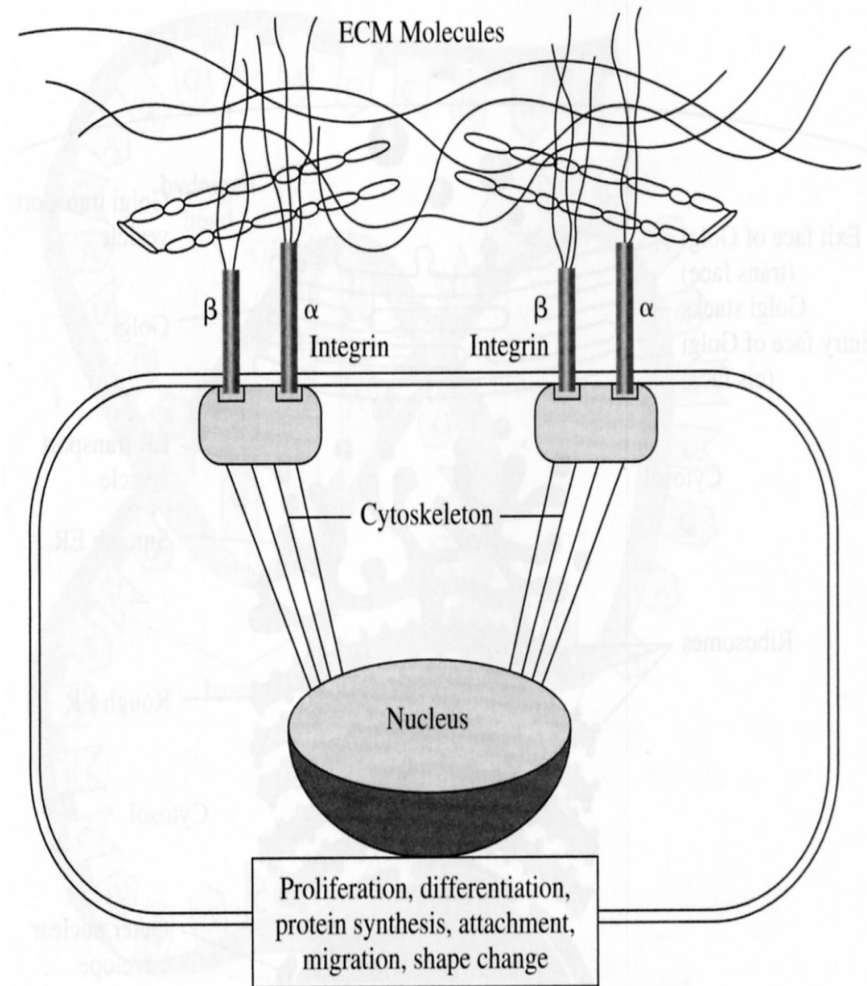
LysoTracker®结构上由一个荧光基团和相连的弱碱基构成，可自由穿过细胞膜，一般聚集在球形细胞器上，适用于观察溶酶体内部生物合成及相关发病机理。**LysoTracker中性pH下仅仅发生部分质子化，因此该探针标记细胞器的原理可能与其完全质子化并滞留在细胞器膜上有关。**

本品LysoTracker® Red DND-99为红色荧光标记的溶酶体探针，具有577/590 nm的最大激发/发射波长。本品以溶于无水DMSO的1mM储存液形式提供。

Extracellular matrix (ECM)

Interactions between a cell and its environment can result in cell **spreading, migration, communication, differentiation** and **activation**. This is called “**outside-in**” signaling.

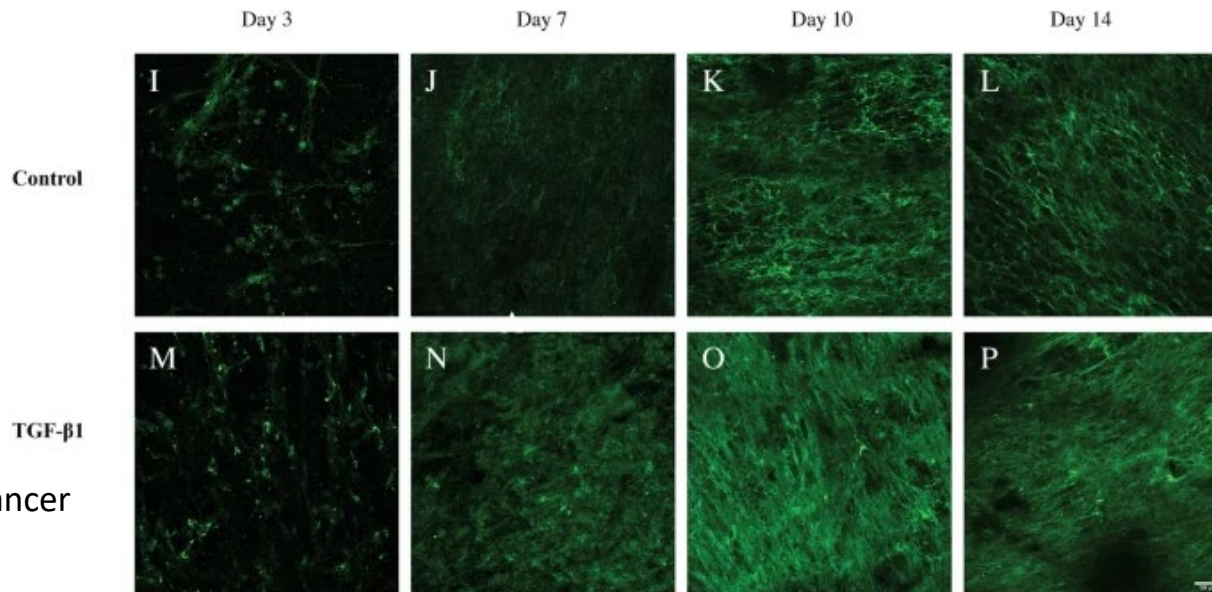
Conversely, a cell may secrete molecules or rearrange contacts to alter the ECM. This is called “**Inside-out**” signaling



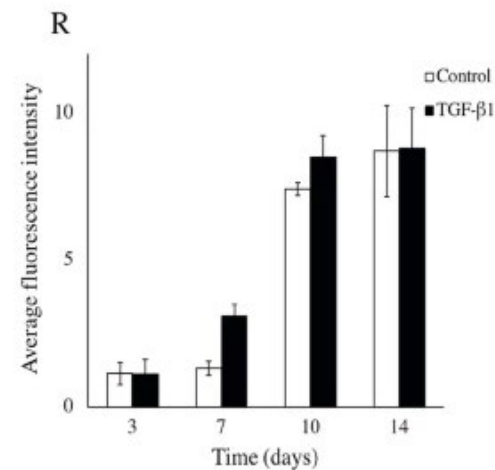
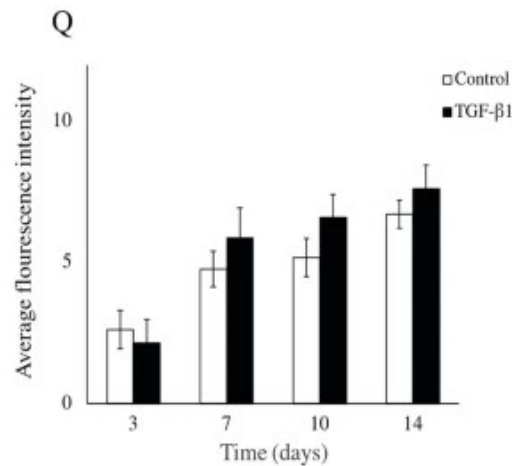
Major compositions of ECM?

Integrin: 整合素；整联蛋白；整合蛋白

JIMT-1/HDF



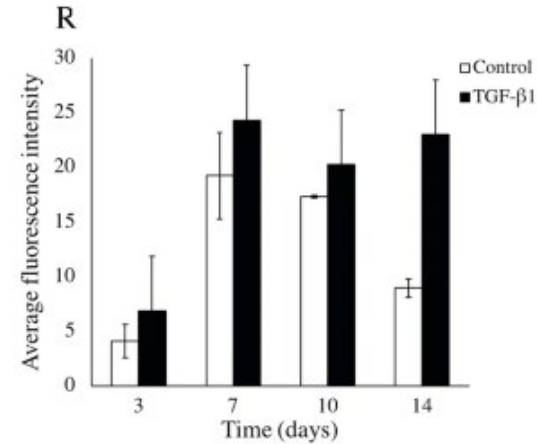
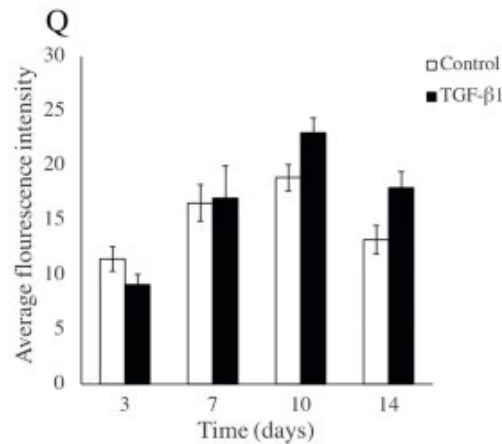
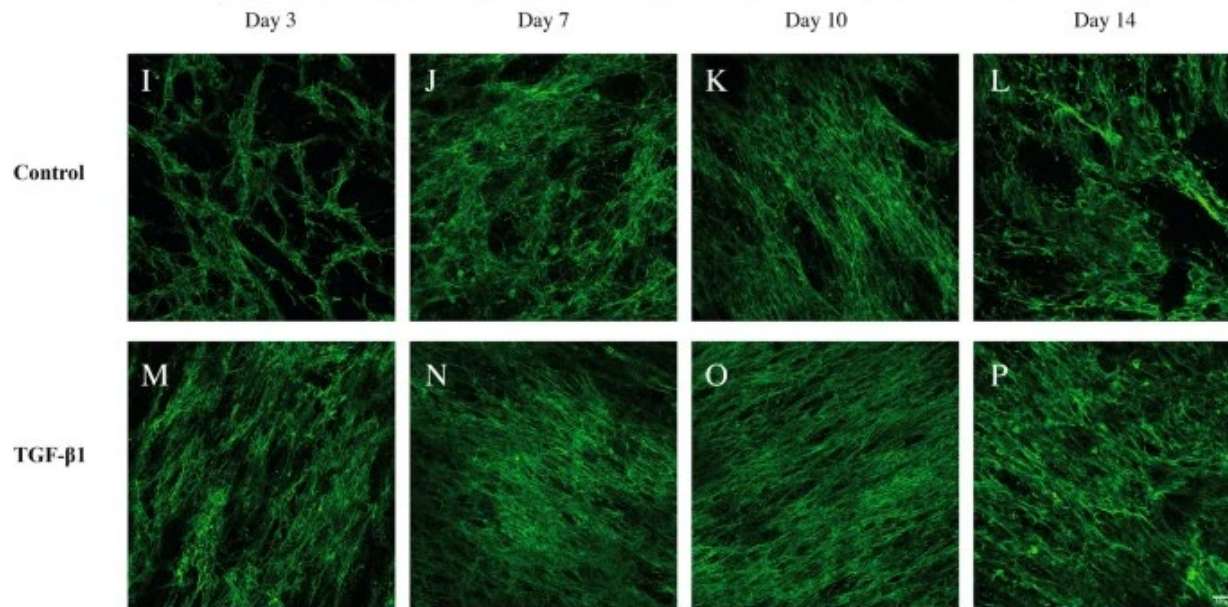
(transforming growth factor- β , TGF- β)是属于一组新近发现的调节细胞生长和分化的TGF- β 超家族



Scientific Reports volume 11, Article number: 6655 (2021)

Collagen I deposition, localization, and quantification in decellularized scaffolds of HDF and **JIMT-1/HDF 3D cultures**. (A–P) Single confocal plane images of collagen I deposition taken in the centre of decellularized 3D cultures incubated in the absence (control) or presence of 5 ng/ml TGF- β 1.

JIMT-1/HDF



Fibronectin deposition, localization, and quantification in decellularized scaffolds of HDF and JIMT-1/HDF 3D cultures

Integrins

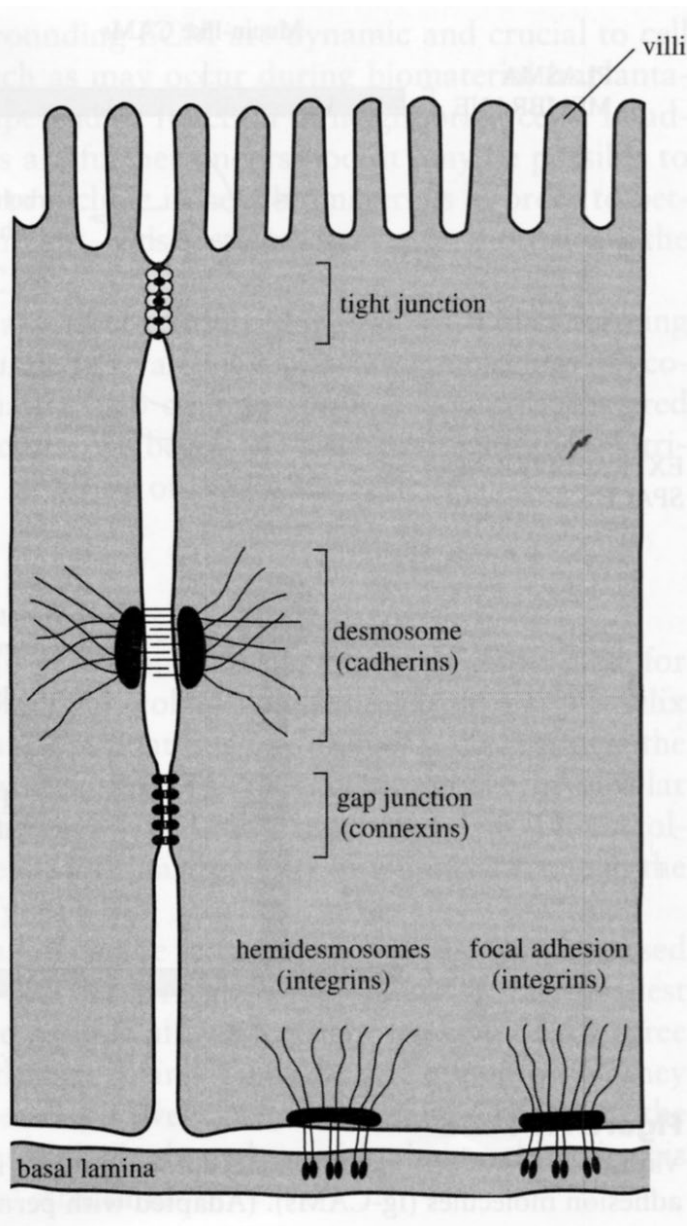
Integrins are transmembrane receptors that facilitate cell-extracellular matrix (ECM) adhesion.

Upon ligand binding, integrins activate signal transduction pathways that mediate cellular signals such as regulation of the cell cycle, organization of the intracellular cytoskeleton, and movement of new receptors to the cell membrane.[3] The presence of integrins allows rapid and flexible responses to events at the cell surface (e.g. signal platelets to initiate an interaction with coagulation factors).

Several types of integrins exist, and one cell may have multiple different types on its surface. Integrins are found in all animals while integrin-like receptors are found in plant cells.[4]

Integrins work alongside other receptors such as cadherins, the immunoglobulin superfamily cell adhesion molecules, selectins and syndecans, to mediate cell–cell and cell–matrix interaction. Ligands for integrins include fibronectin, vitronectin, collagen and laminin.

Types of cell contacts



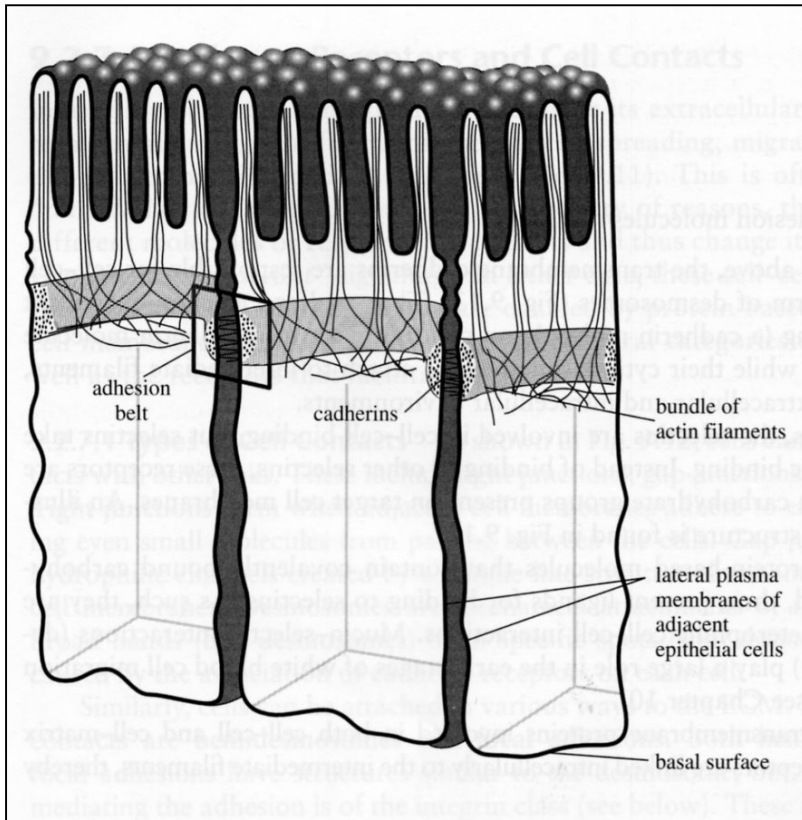
Tight junctions: cells adhere fast to each other

Gap junctions: small hydrophilic channels between cells and membranes

Desmosomes: mechanical attachment between two cells. 胞桥体；胞桥体接合；含较少胞桥体

Cells can attach to ECM via **hemidesmosomes** (半桥粒；半胞桥小体) or **focal adhesion**. These form strong adhesion to the ECM

Cell membrane receptors & ligands



Location of cadherins in epithelial cells. They link to each other to bind cells, and their cytoplasmic regions attach to intermediate filaments linking the ECM to the intracellular environment

Cellular interactions are facilitated by cell membrane receptors, each of which is specific for a small range of target molecules, or **ligands**.

Common receptor molecules are:

Cadherins (钙粘素): responsible for desmosomes; a cadherin molecule on one cell binds to a cadherin on another cell. This is *homophilic* binding

Selectins (选择素): selectins are like cadherins, but bind to other types of receptors, or *heterophilic* binding

Mucins (黏蛋白): participate in heterophilic binding to selectins.

Integrins: transmembrane proteins involved in both cell-cell and cell-matrix contacts.

Cell membrane receptors & ligands

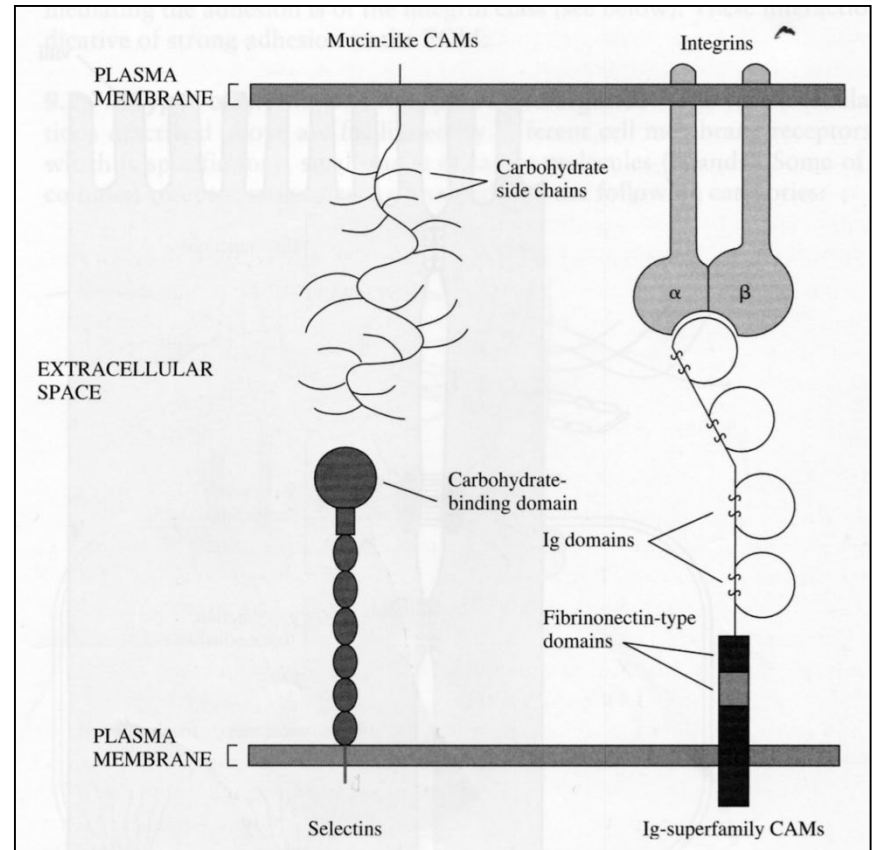
Integrins have two distinct α and β subunits, and are called heterodimers.

Variations in the composition of the α and β chains results in selective adhesion to different ligands

Other **cell adhesion molecules (CAMs)** comprise a large group of membrane proteins that mediate cell-cell interactions via both homophilic and heterophilic binding (嗜异性).

An example of these other CAMs is the immunoglobulin (Ig) family in the picture at right.

The receptors described consist mainly of protein with a small attached carbohydrate (sugar). Similar molecules with small protein and large sugar content are called **proteoglycans**.

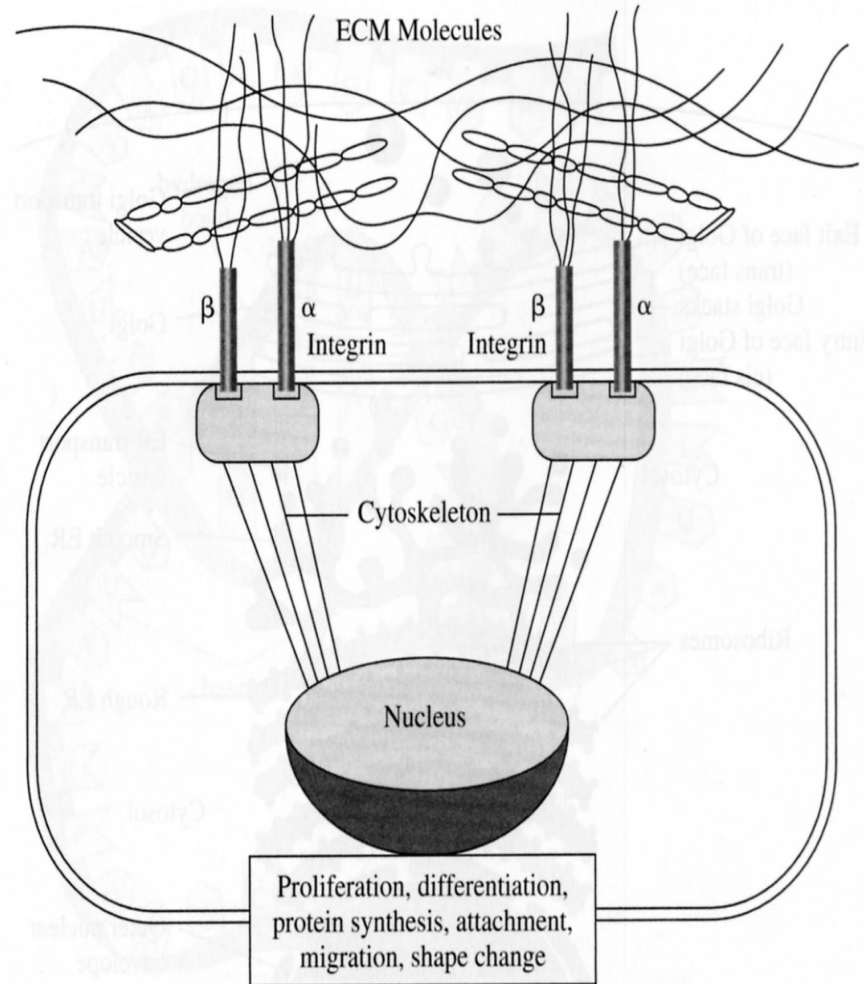


Various types of cell membrane receptors: mucins, integrins, selectins, and Ig-cell adhesion molecules (Ig-CAMs).

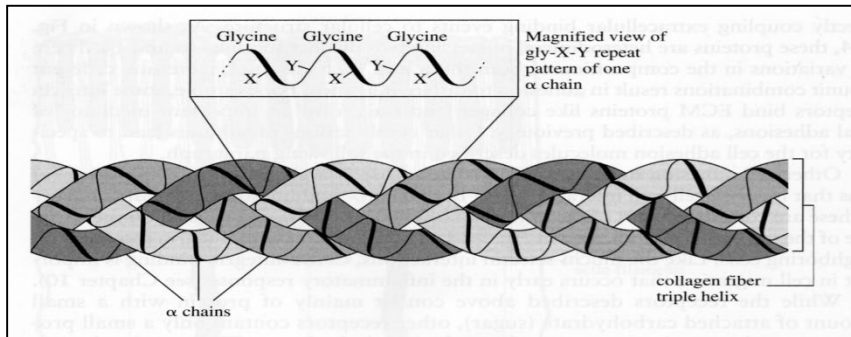
Extracellular Matrix – ECM Environment

Understanding the cell-ECM interaction and response is key to designing new biomaterials

The ECM may be thought of as a fiber reinforced composite with fibers made of collagen or elastin, and a matrix made of glycoproteins and proteoglycans



Collagen



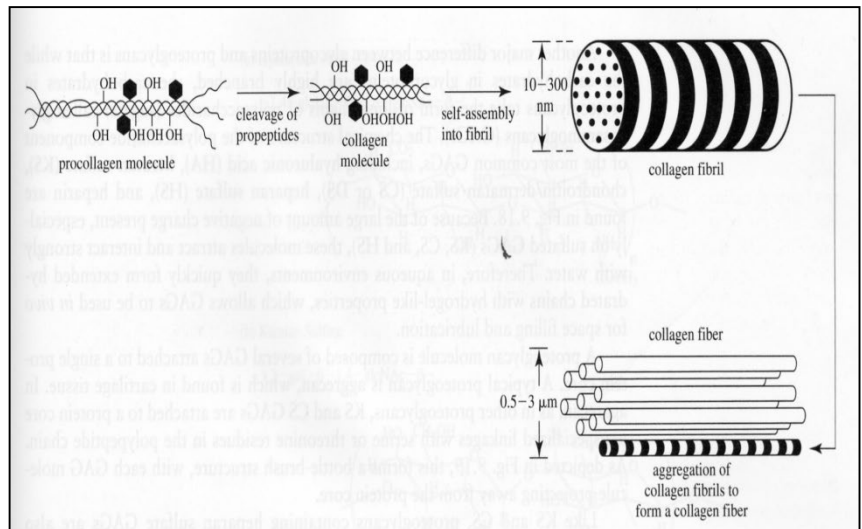
Collagen is the most abundant protein in mammals. It is responsible for tissue tensile strength.

Collagen is made up of α -polypeptides, or amino acids in a gly-x-y pattern (above). Gly is a small molecule resulting in tight packing

X and Y molecules are often proline and hydroxyproline

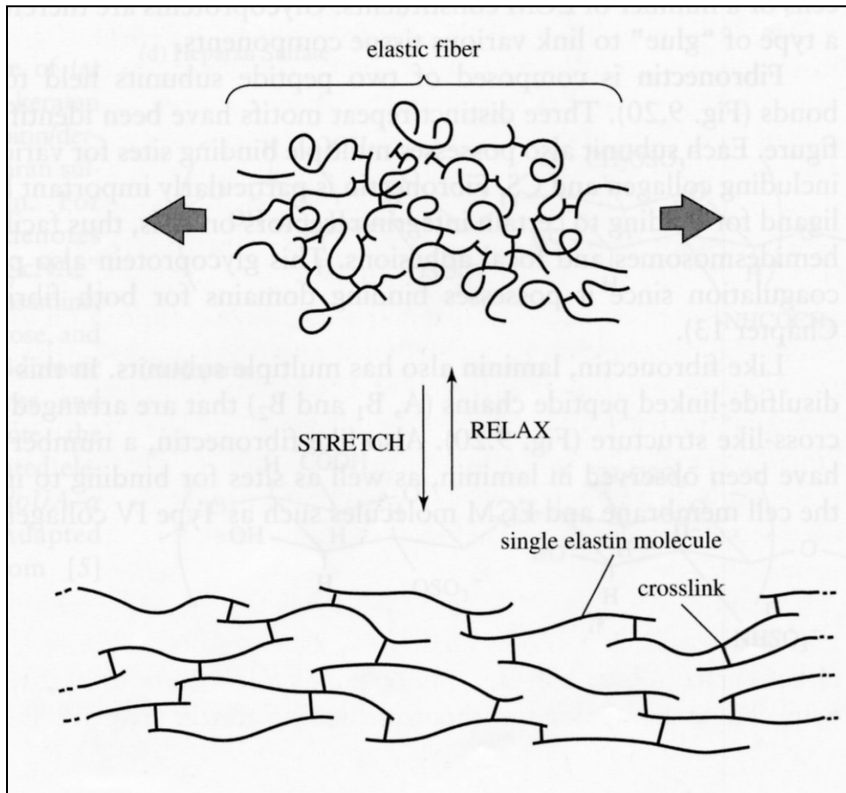
Cell excretes procollagen molecules that self-assemble into fibers (right)

Upon secretion of procollagen from the cell into the ECM, small peptide sequences are cleaved to form the molecule to allow for more efficient packing of collagen molecules into fibrils (10-300 nm). Individual fibers are then assembled into larger fibers (0.5 – 3 μ m diameter)



Properties of collagen may be manipulated by controlling crosslinking of molecules

Elastin



Elastin is responsible for the resiliency and elasticity of the ECM.

Elastin is made up of 85% hydrophobic amino acids

When relaxed, elastin molecules coil up. When a tensile load is applied they unfold into long chains. The chains are crosslinked to adjacent elastin molecules

proteoglycans and glycoproteins

Additional ECM molecules include **proteoglycans** and **glycoproteins**.

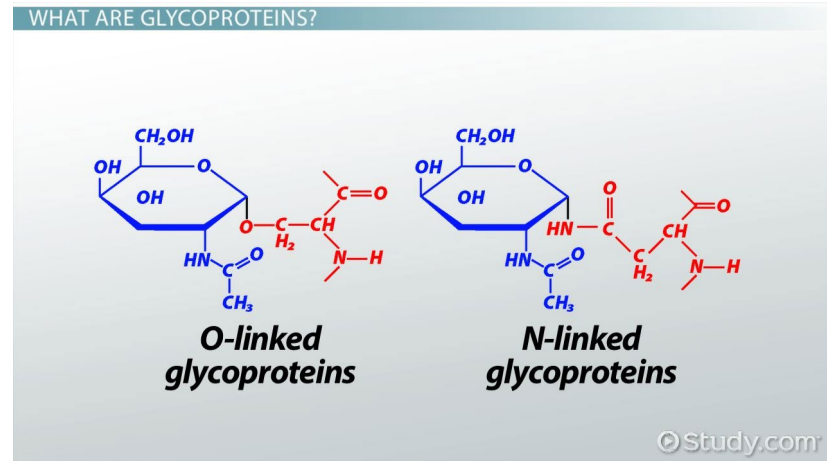
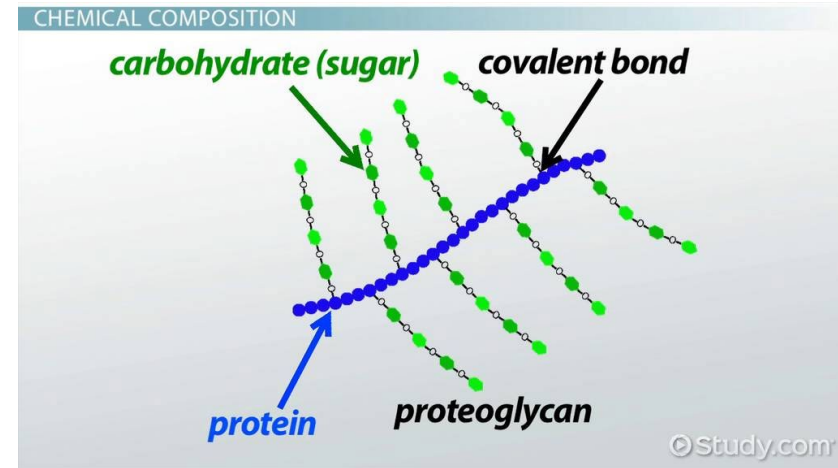
These molecules attract and interact strongly with water.

Carbohydrates in **proteoglycans** form long chains of polysaccharides called glycosaminoglycans (GAGs)

Proteoglycans have several GAGs attached to a protein core.

An aqueous environment is favorable to transport and store bioactive molecules.

The key difference between proteoglycans and glycoproteins is that the proteoglycans have long unbranched chains with disaccharide units as repeating structures while **the glycoproteins have short highly branched glycan chains with no repeating units**.



Proteoglycans: 蛋白多糖 Glycoproteins: 糖蛋白

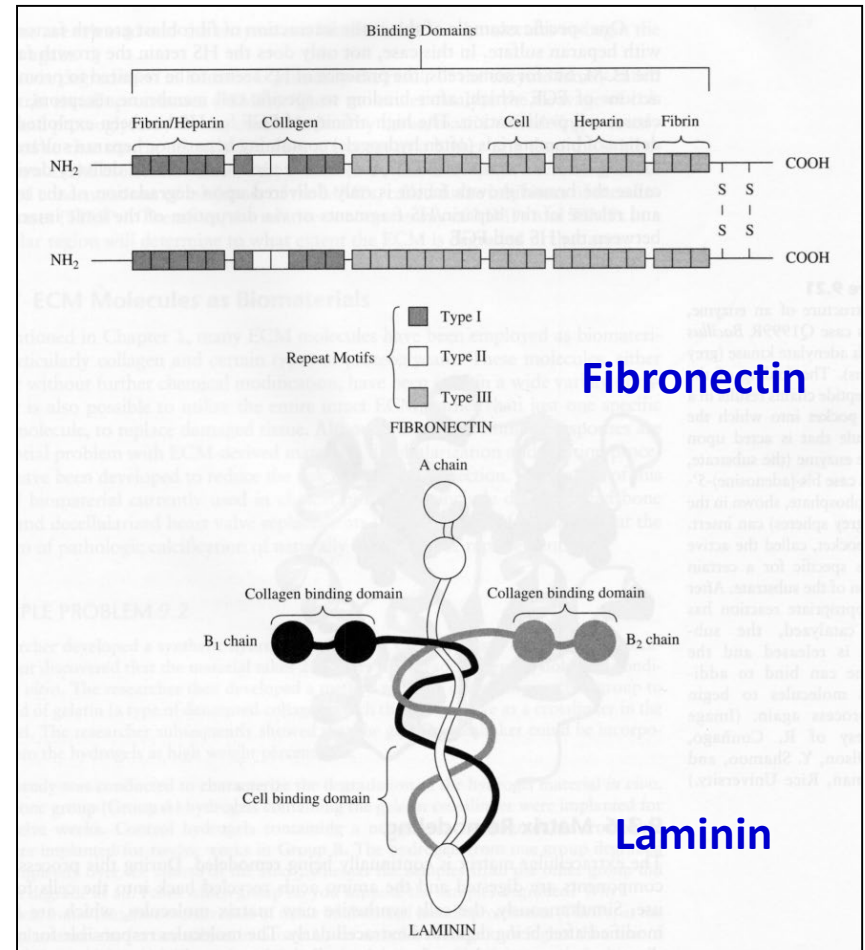
proteoglycans and glycoproteins

Two molecules that represent **glycoproteins** are **fibronectin** and **laminin**.

Each consists of peptide subunits held together by **disulfide bonds** and contain many sites for binding to various ECM molecules.

Laminin (right) consists of three disulfide linked peptide chains in a loosely woven structure with multiple binding sites

Glycoproteins are important in blood clotting (coagulation) since they possess binding domains for heparin and fibrin



Cell-Environment Interactions that Affect Cellular Functions

Interactions between cells and cell/ECM can alter cells function and affect gene expression in the nucleus

Alterations in gene expression affect four major functions:

1. Cell viability
2. Proliferation
3. Differentiation
4. Protein synthesis and communication

Necrosis: cell death from membrane permeability and enzyme leakage, leading to disintegration

Changes in ECM can cause cell death via **necrosis** or **apoptosis** due to changes in the local chemistry (e.g. pH), factors that attach to cell membrane leading to cell death.

Apoptosis: cell shrinkage followed by fragmentation into vesicles containing small groups of organelles. No inflammatory response